International Journal of Sediment Research xxx (xxxx) xxx

Contents lists available at ScienceDirect

International Journal of Sediment Research

journal homepage: www.elsevier.com/locate/ijsrc



Original Research

Long-term impact of the Guxian Reservoir on water and sediment regulation in the middle reaches of the Yellow River Basin

Jieyu Li [a], Lei Li [b], Donglin Li [c], Gang Liu [d], Guoqing Sang [a], Yuanjian Wang [c, e, *]

[a] School of Water Conservancy and Environment, University of Jinan, Jinan 250022, China
[b] Jiangsu Hehai Engineering Technology Co., Ltd., Nanjing 210098, China
[c] Yellow River Institute of Hydraulic Research, YRCC, Zhengzhou 450003, China
[d] School of Water Conservancy and Transportation, Zhengzhou University, Zhengzhou 450001, China
[e] Key Laboratory of Yellow River, MWR, Zhengzhou 450003, China

ARTICLE INFO

Article history:
Received 16 March 2026
Received in revised form
1 September 2026
Accepted 8 October 2026
Available online xxx

Keywords:
Yellow river
Multi-objective
Multi-reservoir system
Water and sediment regulation
Regulatory effect

ABSTRACT

The Yellow River is known for its extremely high sediment concentrations, necessitating effective water and sediment regulation to balance runoff and sediment. Revealing the multiple regulatory effects of the under-construction Guxian (GX) Reservoir is essential for enhancing synergy of the regulation system in the middle reaches. In this study, a long-series multi-objective synergistic water and sediment regulation (LMSWSR) model was established for GX, Sanmenxia (SMX), and Xiaolangdi (XLD) reservoirs. Multiple objectives were considered, including sediment reduction, water supply, power generation, and ecological protection. The regulation performance during the first decade of operation was evaluated under high-, normal-, and low-flow series. The results indicate that the GX Reservoir has positive effects on reservoir and channel scouring, power generation, and downstream ecology. Specifically, the GX Reservoir significantly increases the inflow of the XLD Reservoir in July across all hydrological series, thereby enhancing its hydrodynamic force of water and sediment regulation. The sediment releasing ratios have a relatively minor impact on the Tongguan Elevation. Compared to the scenario without the GX Reservoir, the 15% sediment releasing ratio scheme lowers the Tongguan elevation by 0.53 m, 0.37 m, and 0.56 m under the high-, normal-, and low-flow series, respectively. Furthermore, an analysis of different releasing ratios reveals that a 15% ratio achieves optimal sediment transport efficiency under high-flow conditions. Conversely, under normal- and low-flow conditions, lowering the ratio to 10% creates a more favorable hydrodynamic process, maximizing the sediment discharge into the sea. This study contributes to the advancement of integrated water and sediment regulation strategies for cascade reservoirs in the Yellow River Basin.

© 2026 International Research and Training Centre on Erosion and Sedimentation. Publishing services by Elsevier B.V. on behalf of KeAi Communications Co. Ltd. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).

1. Introduction

By 2019, approximately 60,000 large-scale dams had been constructed worldwide, with a total storage capacity of 7720.2 km^3 (Wang et al., 2021). In major river basins, integrated regulation by cascade reservoirs plays a critical role in adjusting the runoff-sediment relationship, decelerating sedimentation in reservoirs and channels, ensuring the efficient utilization of water resources, promoting stable socio-economic development, and maintaining a healthy ecological environment. However, integrated regulation by cascade reservoirs may also generate negative effects (Belletti et al., 2020; Li et al., 2024a). Recently, the cumulative effects of these regulations have attracted increased attention.

The concept of cumulative effects originates from the field of environmental impact assessment and reflects the incremental impact of an activity on the environment when combined with the effects of past, present, and foreseeable future activities (Canter & Kamath, 1995). With the construction of cascade reservoirs in river basins, a common phenomenon associated with the regulation of reservoirs is the substantial alteration of natural runoff processes

* Corresponding author. Yellow River Institute of Hydraulic Research, YRCC, Zhengzhou 450003, China.
E-mail address: wangyuanjian_yrcc@aliyun.com (Y. Wang).

Peer review under the responsibility of editorial board of International Journal of Sediment Research

https://doi.org/10.1016/j.ijsrc.2026.10.001

1001-6279/© 2026 International Research and Training Centre on Erosion and Sedimentation. Publishing services by Elsevier B.V. on behalf of KeAi Communications Co. Ltd. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).

Please cite this article as: Li, J et al., Long-term impact of the Guxian Reservoir on water and sediment regulation in the middle reaches of the Yellow River Basin, International Journal of Sediment Research, https://doi.org/10.1016/j.ijsrc.2026.10.001

and considerable interception of sediment. In large river basins, where reservoirs are densely distributed, the influence of individual reservoirs on water and sediment transport processes inevitably accumulates (Zhou et al., 2020). Research on environmental cumulative effects has primarily applied theoretical approaches, such as GIS-based methods, system dynamics, scenario analysis, matrix methods, and model simulations, to analyze sources of impact, cumulative processes, spatial accumulation, and temporal accumulation (Li et al., 2024c). Data analysis and hydrological modeling have been applied to evaluate variations of hydrological elements under the influence of cascade reservoirs (Habets et al., 2018). The cumulative effects of cascade reservoirs on hydrological elements often exceed the sum of individual reservoir effects (Dos Santos et al., 2018; Wang et al., 2018). Timpe and Kaplan (2017) statistically analyzed hydrologic indicators of the Amazon River and found that cumulative effects of multiple large dams on the frequency and duration of high- and low-pulse flows were often larger than effects of a single dam. Guo et al. (2020, 2024) continuously studied the impact of cascade reservoirs on the streamflow and sediment load changes in the Yangtze River, proposed a reservoir storage capacity index, and discovered that a threshold of more than 4% might lead to a significant reduction in sediment load. Furthermore, flood retention amount was the major factor causing siltation in the Three Gorges Reservoir. A few studies have employed hydrodynamic models to simulate the temporal cumulative trends of sediment deposition in cascade reservoirs (Wang et al., 2023).

The Yellow River is one of the most complex and challenging rivers due to its extremely high sediment concentration. As a key technical approach to adjust the runoff-sediment relationship, integrated water and sediment regulation by cascade reservoirs has played a crucial role in flood control, sediment reduction, water supply, power generation, and ecological protection (Jiang et al., 2023). The Sanmenxia (SMX) and Xiaolangdi (XLD) are main reservoirs located in the middle reaches of the Yellow River. Current assessments of cumulative effects associated with regulation by reservoirs have been mainly focused on siltation and erosion of reservoirs and channels, and ecological environmental impacts.

Research on siltation and erosion of reservoirs and channels has primarily focused on variations in Tongguan Elevation and on sedimentation processes in reservoirs and downstream channels of the Yellow River. The Tongguan (TG) Station is the inflow control station of the SMX Reservoir. During the reservoir's early operation stage, underestimated sediment inflow led to rapid reservoir siltation and a sharp rise of the Tongguan Elevation, thereby compromising the flood control safety of the Wei River Plain (Wu et al., 2004). Two structural modifications and three operational changes at the SMX Reservoir substantially influenced erosion within the reservoir and helped stabilize the Tongguan Elevation (Chen et al., 2017; Wang et al., 2005). The cumulative sediment volume in the reservoir area below TG station remains 2.8 billion m^3, which preserves the long-term effective storage of the SMX Reservoir. Nevertheless, persistently high Tongguan Elevation has not been fundamentally alleviated. Since the XLD Reservoir entered service, integrated water and sediment regulation by the reservoirs in the middle reaches of the Yellow River has been conducted annually. This regulation has substantially reduced the sedimentation rate in the XLD Reservoir relative to design expectations, while producing continuous downstream channel scouring (Xia et al., 2025). From 2000 to 2022, the downstream channel was scoured by a cumulative 2.402 billion m^3, and the minimum bankfull discharge reached 4700 m^3/s (Jiang et al., 2023; Liu et al., 2021; Wu et al., 2023). These effects have effectively mitigated sedimentation in the downstream channels and improved the flow capacity of the main channel.

Regarding ecological environmental impacts, integrated regulation of cascade reservoirs has reshaped water and sediment transport, substantially improving instream flow reliability, enhancing habitat stability, and promoting the recovery of freshwater fish biodiversity. Ecological water replenishment in the Yellow River Delta has optimized ecosystem structure in the lower reaches and estuary, expanded delta wetlands and water bodies, and markedly increased the habitat index (Jiang et al., 2023). Numerous studies have examined specific ecological issues. For instance, Hou et al. (2021) investigated how water and sediment regulation altered sediment grain size and nutrient content in the Yellow River and reported that channel scouring reduced total organic carbon and total phosphorus in sediments by approximately 40%. Zhang et al. (2021a) analyzed the effects of different regulation schemes on phytoplankton communities in the Yellow River estuary and supplied a basis for predicting phytoplankton responses to increased hydrodynamics, nutrient loading, and turbidity. Mu et al. (2024) used isotopic analysis to examine nitrogen cycling and its drivers during the release period of the XLD Reservoir, finding that nitrogen cycling in cascade reservoirs is more complex than in rivers and thereby deepening understanding of anthropogenic impacts on nitrogen dynamics in river−reservoir systems.

Although cascade reservoirs in the middle reaches of the Yellow River have been crucial for water and sediment control, the system's backbone reservoirs remain incomplete, and overall coordination still needs improvement (Li et al., 2024b). The construction of the Guxian (GX) Reservoir in the middle reaches of the Yellow River will inject new momentum into the basin's water and sediment regulation system (Zhang et al., 2021b). In recent years, several studies have examined the regulation performance expected after the completion of the GX Reservoir. Sun et al. (2022) examined the flow structure at the GX Reservoir's confluence with its tributaries and emphasized its importance for sediment transport and riverbed evolution at reservoir−tributary junctions, offering guidance for sediment discharge regulation. The commissioning of the GX Reservoir will markedly enhance the water and sediment regulation capacity of the XLD Reservoir. Simultaneously, by intercepting sediment and regulating water and sediment flows, the GX Reservoir can mitigate unfavorable water and sediment conditions in the channels of the middle and lower reaches, induce continuous scouring, and substantially lower the Tongguan Elevation. In addition to these regulation functions, the GX Reservoir yields other benefits (Wan et al., 2013). Jin et al. (2024) assessed its multi-objective contributions to the middle and lower Yellow River and found that the GX Reservoir can substantially enhance water supply, power generation, water and sediment regulation, and ecological protection.

Although prior research has evaluated the GX Reservoir's regulatory performance after construction, most studies have examined either a single regulatory effect or the enhancement of multiple benefits. Comprehensive assessments of the long-term integrated regulatory performance of the joint operation of GX-SMX-XLD reservoirs remain scarce. In this study, a long-series multi-objective synergistic water and sediment regulation (LMSWSR) model of GX-SMX-XLD reservoirs was established. The water and sediment series of high-, normal-, and low-flows were designed, and scenarios with and without the GX Reservoir were compared. The impacts of the GX Reservoir on sediment reduction, water supply, power generation, and ecology were analyzed. Furthermore, performance evaluation indicators were developed to assess their overall influence on enhancing the hydrodynamic force of water and sediment regulation in the XLD Reservoir, reducing Tongguan Elevation through scouring, and improving runoff utilization efficiency. The findings could provide technical

Please cite this article as: Li, J et al., Long-term impact of the Guxian Reservoir on water and sediment regulation in the middle reaches of the Yellow River Basin, International Journal of Sediment Research, https://doi.org/10.1016/j.ijsrc.2026.10.001

support for optimizing the integrated regulation strategies of the GX-SMX-XLD reservoirs and promoting sustainable development in the middle and lower reaches of the Yellow River.

2. Research area and data

2.1. Research area

In this study, the middle and lower reaches of the Yellow River Basin, extending from the GX Reservoir to the estuary, approximately 1340 km (Fig. 1), were selected as the research area. The major tributaries in this region include the Wuding River, Wei River, Yiluo River, Fen River, Qin River, and Dawen River. Tributaries in the middle reaches mostly flow through the Loess Plateau, where severe soil erosion is prevalent, making them prone to flooding and high sediment loads. The 800 km stretch of the lower reaches experienced significant sedimentation, with the riverbed elevated 4–6 m above the ground surface, forming a "suspended river". According to the *Comprehensive Planning for the Yellow River Basin (2012–2030)* (Yellow River Conservancy Commission (YRCC), 2013), the backbone reservoirs of the water and sediment regulation subsystem in the middle reaches include the SMX and XLD reservoirs, as well as the under-construction GX Reservoir and the planned Qikou Reservoir. Their primary functions are flood prevention and sediment reduction, while also considering additional beneficial uses.

2.2. Data sources

This study considered the GX Reservoir, the SMX Reservoir, and the XLD Reservoir, as well as the Xiaobei mainstream and the downstream channel of the Yellow River. The Xiaobei mainstream is the section between the Longmen (LM) Station and the SMX Reservoir. The downstream channel is below the Huayuankou (HYK) station downstream of the XLD Reservoir. Table 1 summarizes the basic characteristics of the three reservoirs. Inflows from three tributaries in the middle reaches, such as the Wei River, the Beiluo River, and the Fen River, are incorporated. Their control stations involve the Huaxian (HX) Station, the Zhuangtou (ZT) Station, and the Hejin (HJ) Station, respectively. For the GX Reservoir, the inflow station on the mainstream is the LM Station, located downstream of the dam site.

Table 1
Characteristics of the reservoirs.

Characteristics	GX	SMX	XLD
Dead water level (m)	588	300	230
Flood control water level (m)	617	305	230/248
Normal high water level (m)	627	318	270
Sediment retention capacity (10^8 m^3)	93.42	/	75.5
Water and sediment regulation capacity (10^8 m^3)	20	5.72	51
Maximum power discharge (m^3/s)	1716	1,500	1776
Installed capacity (MW)	2,100	450	1800

According to the *Feasibility Study Report of the Guxian Water Conservancy Project in the Yellow River Basin*, the early stage for sediment retention lasts approximately 10 years. To assess the long-term impacts after the operation of the GX Reservoir, a 10-year series of optimized regulation was conducted, with non-flood-season regulation on a monthly scale and flood-season regulation on a 10-day scale. The water and sediment series of high-, normal-, and low-flow series were also designed. Considering that the water and sediment conditions in China generally follow a P-III distribution, the runoff data from a 10-year sliding average (2001–2020) at four stations in the middle reaches were fitted to a P-III distribution. Values below 37.5% were classified as high-flow series, values between 37.5% and 62.5% as normal-flow series, and values above 62.5% as low-flow series. Hence, 2011–2020 was identified as the high-flow series, 2006–2015 as the normal-flow series, and 2002–2011 as the low-flow series. Sediment inflow data corresponding to these years were applied (Fig. 2). The inflow, outflow, and sediment transport data were sourced from the hydrological yearbook of the Yellow River Basin (Ministry of Water Resources of China, 2001–2020), whereas the water supply data were obtained from the *Yellow River Water Resources Bulletin* (YRCC, 2001–2020).

In the proposed model, ecological suitable flow satisfaction rates were primarily evaluated for the Xiaobei mainstream and downstream channel of the Yellow River. The ecological suitable flows, ecological basic flows, and bankfull discharges of the Xiaobei mainstream and downstream channel are listed in Table 2.

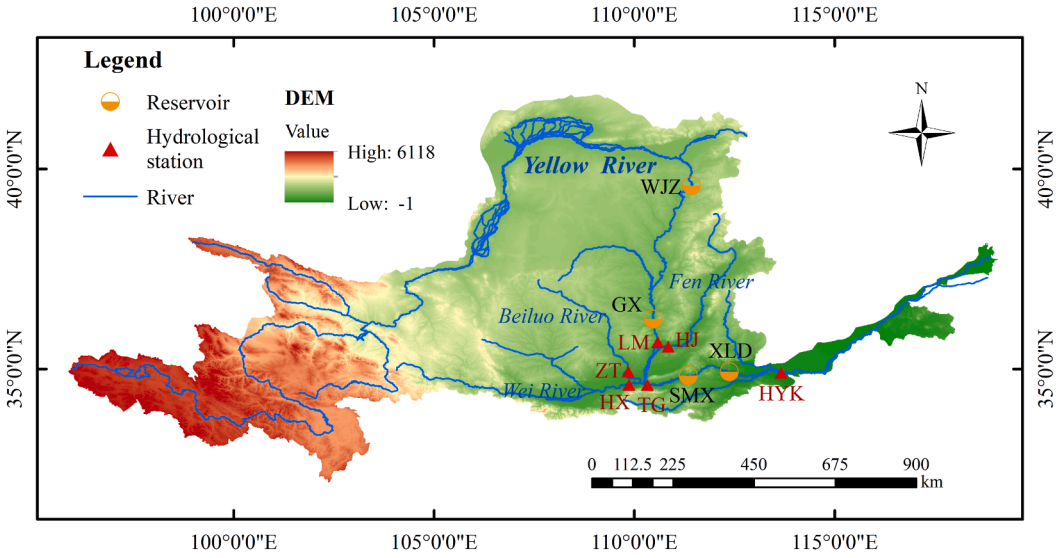


Fig. 1. Sketch map of the research area.

Please cite this article as: Li, J et al., Long-term impact of the Guxian Reservoir on water and sediment regulation in the middle reaches of the Yellow River Basin, International Journal of Sediment Research, https://doi.org/10.1016/j.ijsrc.2026.10.001

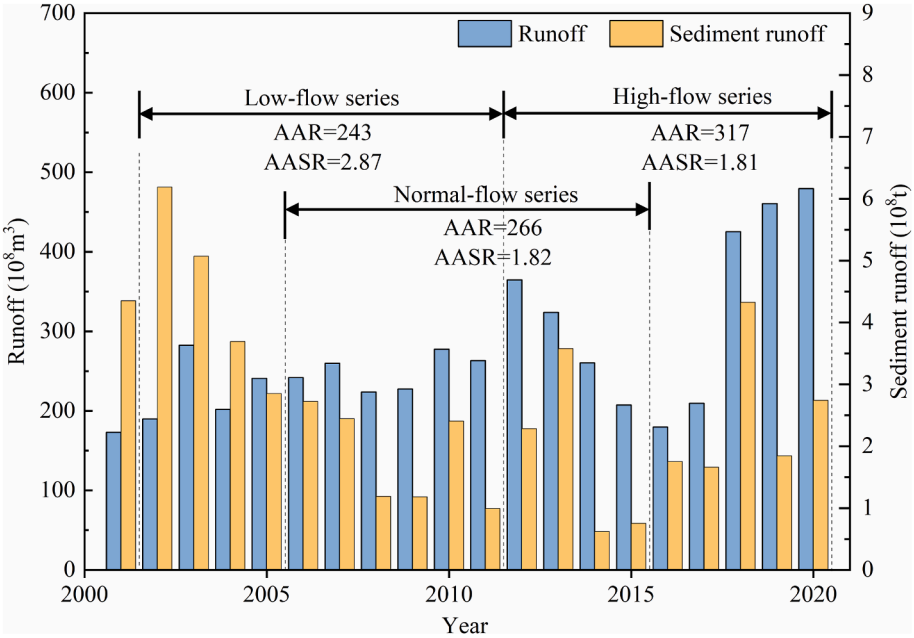


Fig. 2. Annual runoff and sediment runoff of four stations in the middle reaches (AAR represents the average annual runoff and AASR represents the average annual sediment runoff.).

Table 2
Ecological flow and bankfull discharges of channels.

Discharge (m^3/s)	Ecological suitable flow		Ecological basic flow		Bankfull discharge
	Nov.–Mar.	Apr.–Oct.	Apr.–Jun.	Jul.–Mar.	
Xiaobei mainstream	240	240	240	180	4000
Downstream channel	400	800	200	200	4800

3. Methodology

The framework of this paper is illustrated in Fig. 3. First, the LMSWSR model of GX-SMX-XLD reservoirs is constructed. Then, the Non-dominated Sorting Genetic Algorithm II (NSGA-II) is applied to obtain a Pareto set, and a multiple attribute decision-making method is employed to identify the optimal solution. Finally, the evaluation indicators for regulation performance are established to compare the effectiveness with and without the GX reservoir.

3.1. Establishment of the GX-SMX-XLD long-series multi-objective synergistic water and sediment regulation (LMSWSR) model

3.1.1. Objective function

(1) Maximizing the reservoir and channel erosion quantity

$$W_s = \max \sum_{t=1}^{T} \left(\left(\sum_{i=1}^{M} S_{Ri,t} \right) + \left(\sum_{j=1}^{N} S_{Cj,t} \right) \right) \quad (1)$$

where W_s is the total erosion quantity of reservoirs and channels (t); $S_{Ri,t}$ is the erosion quantity of the ith reservoir in the tth period (t); $S_{Cj,t}$ is the erosion quantity of the jth channel in the tth period (t); T is the operation period; M is the number of reservoirs; and N is the number of channels.

For the sediment retention strategy of the GX Reservoir, no sediment was released during the non-flood season, whereas the sediment was released at a ratio of 15% during the flood season. This 15% sediment releasing ratio was derived from an early-stage sediment retention scheme as documented in the *Feasibility Study Report of the Guxian Water Conservancy Project in the Yellow River Basin* (Li et al., 2021). In order to demonstrate the impact of different sediment releasing ratios on the regulation effect of GX Reservoir, the regulation performance under three sediment releasing ratios of 10%, 15%, and 20% during the flood season of GX Reservoir was compared in Section 4.3.

When calculating the sedimentation and erosion volumes of the Xiaobei mainstream, SMX Reservoir, XLD Reservoir, and the downstream channel, empirical equations were obtained by calibrating based on historical data from 2001 to 2020. The details are as follows.

Owing to the relatively small storage capacity of SMX Reservoir, its sediment release during the flood season followed the sediment release pattern of channels. In contrast, the XLD Reservoir had a relatively large storage capacity. When the water level is high and the reservoir remains in an impoundment state during the flood season, only a small amount of sediment can be released. However, as the water level decreased to a relatively low level, sediment release increased substantially. Based on the variation in the water levels during sediment release, the process can be divided into two modes: high-water-level and low-water-level sediment release. The sediment release Equations for SMX and XLD reservoirs are calculated according to the equations presented in Liu et al. (2025).

Sediment release of the SMX Reservoir is described by

Please cite this article as: Li, J et al., Long-term impact of the Guxian Reservoir on water and sediment regulation in the middle reaches of the Yellow River Basin, International Journal of Sediment Research, https://doi.org/10.1016/j.ijsrc.2026.10.001

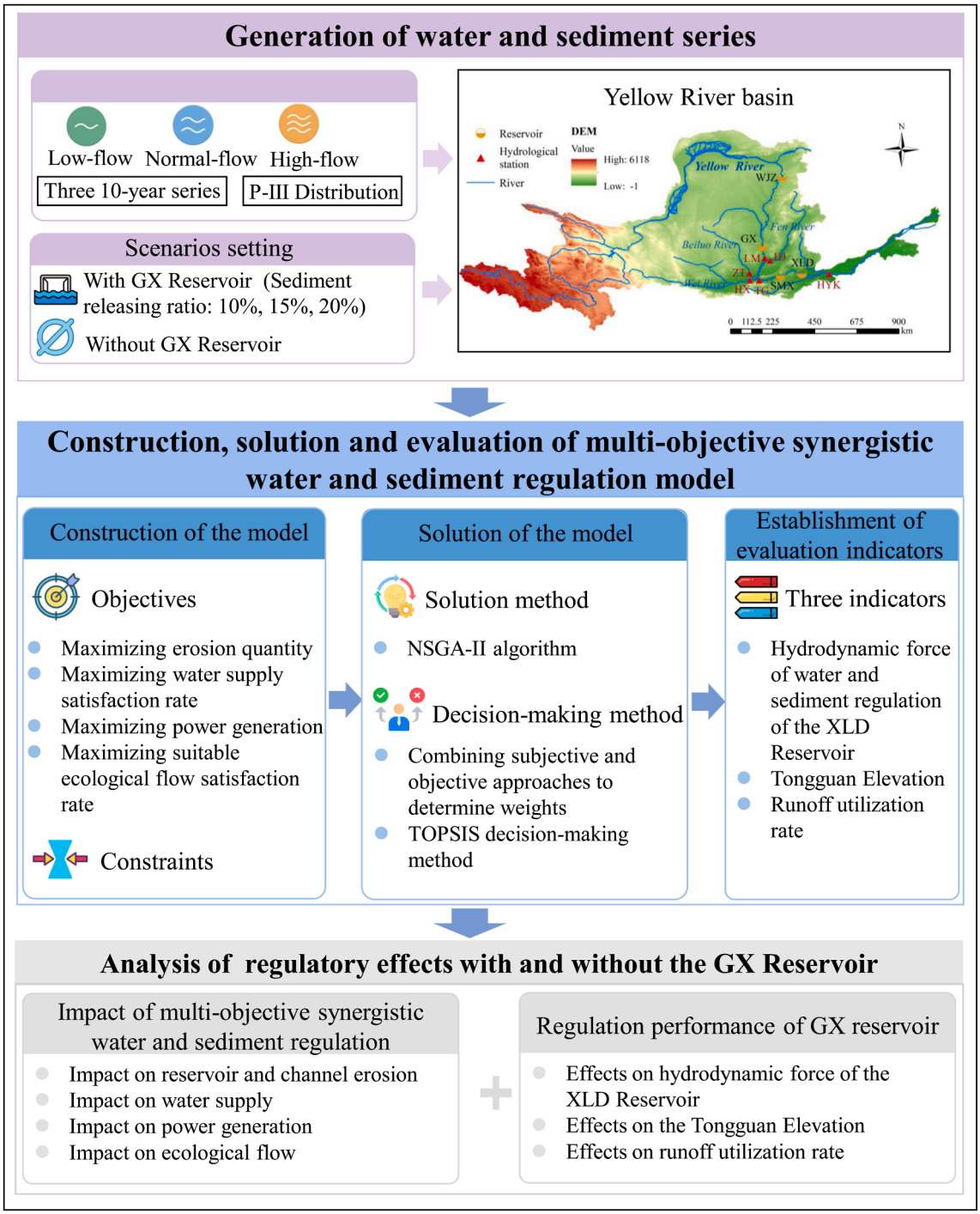


Fig. 3. Framework of this paper.

$$Q_{s,\mathrm{smx},t} = 0.00361 \times Q_{\mathrm{out},\mathrm{smx},t}^{0.941} S_{in,\mathrm{smx},t}^{0.832} \quad (2)$$

where $Q_{s,\mathrm{smx},t}$ is the outbound sediment transport rate of the SMX Reservoir in the tth period (t/s); $Q_{\mathrm{out},\mathrm{smx},t}$ represents the outflow of the SMX Reservoir in the tth period (m^3/s); and $S_{in,\mathrm{smx},t}$ is the inbound sediment concentration of the SMX Reservoir in the tth period (kg/m^3).

High-water-level sediment releasing of the XLD Reservoir is described by

$$\eta = 0.347 \left(\frac{\overline{V}_t}{Q_{out,\mathrm{xld},t}} \right)^{-0.341} \left(\frac{Q_{in,xld,t}}{Q_{out,\mathrm{xld},t}} \right)^{-2.24} S_{in,\mathrm{xld},t}^{-0.475} \quad (3)$$

where η is the sediment releasing ratio; $\overline{V}_t$ is the average storage capacity of the XLD Reservoir in the tth period (m^3); $Q_{in,\mathrm{xld},t}$ is the inflow of the XLD Reservoir in the tth period (m^3/s); $Q_{out,\mathrm{xld},t}$ is the outflow of the XLD Reservoir in the tth period (m^3/s); and $S_{in,\mathrm{xld},t}$ is the inbound sediment concentration of the XLD Reservoir in the tth period (kg/m^3).

Please cite this article as: Li, J et al., Long-term impact of the Guxian Reservoir on water and sediment regulation in the middle reaches of the Yellow River Basin, International Journal of Sediment Research, https://doi.org/10.1016/j.ijsrc.2026.10.001

Low-water-level sediment releasing of the XLD Reservoir can be described by

$$Q_{s,\mathrm{xld},t} = 103.23 \frac{Q_{out,\mathrm{xld},t}^{1.6} J_t^{1.2}}{B^{0.6}} \quad (4)$$

where $Q_{s,\mathrm{xld},t}$ is the outbound sediment transport rate of the XLD Reservoir in the tth period (t/s); J_t is the reach gradient of the XLD Reservoir in the tth period; and B is average width of the XLD Reservoir in the tth period (m).

The erosion and siltation equation for the Xiaobei mainstream during the flood season on a 10-days basis and during the non-flood season on a monthly basis are obtained by calibrating with historical data from 2001 to 2020 at the LM, HX, ZT, HJ, and Tongguan (TG) stations. Assuming that the sediment from tributaries does not deposit in the Xiaobei mainstem, only the sediment of the LM Station experiences deposition within this reach. Therefore, after subtracting the sediment runoff measured at the HX, ZT, and HJ stations of the tributaries from the sediment runoff at the TG station, variable $W_{s,\mathrm{TG},t}$ was obtained. The relationship between $W_{s,\mathrm{TG},t}$ and the runoff and sediment runoff at the LM station is then established, as shown below:

$$W_{s,\mathrm{TG},t} = 0.0015 \times {W_{q,\mathrm{LM},t}}^{1.3692} + 0.972 \times {W_{s,\mathrm{LM},t}}^{1.239} (\text{flood season}) \quad (5)$$

$$W_{s,\mathrm{TG},t} = 0.0062 \times {W_{q,\mathrm{LM},t}}^{0.463} + 1.597 \times {W_{s,\mathrm{LM},t}}^{1.145} (\text{non} - \text{flood season}) \quad (6)$$

where, $W_{s,\mathrm{TG},t}$ is the sediment runoff at the TG station by deducting the sediment runoff from tributaries in the tth period (10^8 t); $W_{q,\mathrm{LM},t}$ is the runoff at LM station in the tth period (10^8 m^3); $W_{s,\mathrm{LM},t}$ is the sediment runoff at LM station in the tth period (10^8 t).

Therefore, the sediment erosion and siltation equation for the Xiaobei mainstream is

$$S_{C1,t} = W_{s,\mathrm{LM},t} - W_{s,\mathrm{TG},t} \quad (7)$$

where, $S_{C1,t}$ is the sediment erosion and siltation amount in the tth period of the Xiaobei mainstream (10^8 t). In the LMSWSR model, the sediment inflow of the SMX reservoir was calculated by subtracting the sediment erosion and siltation amount of the Xiaobei mainstream from the sediment outflow of the GX Reservoir, and then adding the sediment runoff measured at the HX, ZT, and HJ stations of the tributaries.

The erosion and siltation equations for the downstream channel are also divided into two sets of calibration results for flood season on a 10-days basis and non-flood season on a monthly basis. The flood season is further divided into two scenarios: when the XLD Reservoir releases sediment and when it does not release sediment.

In the flood season, if the XLD reservoir releases sediment, the sediment runoff at the Lijin (LJ) station at the estuary of the Yellow River is calculated using the following equation

$$W_{s,\mathrm{LJ},t} = 0.239 \times {W_{q,\mathrm{XLD},t}}^{1.336} + 4.51 \times {W_{s,\mathrm{XLD},t}}^{0.339} \quad (8)$$

In the flood season, if the XLD reservoir does not release sediment, the sediment runoff at the LJ station is calculated using the following equation:

$$W_{s,\mathrm{LJ},t} = 0.0002 \times {W_{q,\mathrm{XLD},t}}^2 + 0.0019 \times W_{q,\mathrm{XLD},t} - 0.0031 \quad (9)$$

In the non-flood season, the sediment runoff at the LJ station is calculated using the following equation

$$W_{s,\mathrm{LJ},t} = 0.0001 \times {W_{q,\mathrm{XLD},t}}^2 - 0.0023 \times W_{q,\mathrm{XLD},t} + 0.0198 \quad (10)$$

where, $W_{s,\mathrm{LJ},t}$ is the sediment runoff at the LJ station in the tth period (10^8 t); $W_{q,\mathrm{XLD},t}$ is the release water volume of the XLD reservoir in the tth period (10^8 m^3); $W_{s,\mathrm{XLD},t}$ is the release sediment amount of the XLD reservoir in the tth period (10^8 t).

Therefore, the sediment erosion and siltation equation for the downstream channel is

$$S_{C2,t} = W_{s,\mathrm{XLD},t} - W_{s,\mathrm{LJ},t} \quad (11)$$

where, $S_{C2,t}$ is the sediment erosion and siltation amount in the tth period of the downstream channel (10^8 t).

(2) Maximizing the water supply satisfaction rate

$$\delta_W = \max \sum_{i=1}^{M} \sum_{t=1}^{T} \left(\frac{W_t}{W_{t,D}} \right) (\text{if } W_t > W_{t,D}, W_t = W_{t,D}) \quad (12)$$

where δ_W is the water supply satisfaction rate of the GX reservoir (%); W_t is the actual water supply volume of the GX reservoir in the tth period (m^3); and $W_{t,D}$ is the design water supply volume of the GX reservoir in the tth period (m^3).

(3) Maximizing power generation of the GX, SMX, and XLD reservoirs

$$E = \max \sum_{i=1}^{M} \sum_{t=1}^{T} k_i Q_{i,t,p} H_{i,t} \Delta t \quad (13)$$

where E is the power generation (kW·h); k_i is the output coefficient of the ith reservoir; $Q_{i,t,p}$ is the power generation flow of the ith reservoir in the tth period (m^3/s); $H_{i,t}$ is the power generation head of the ith reservoir in the tth period (m), and Δt is the time interval.

(4) Maximizing ecological suitable flow satisfaction rate

$$F = \max \sum_{j=1}^{N} \sum_{t=1}^{T} \left(\frac{Q_{j,t,e}}{Q_{j,t,E}} \right) \left(\text{if } Q_{j,t,e} > Q_{j,t,E}, Q_{j,t,e} \Big/ Q_{j,t,E} = 1 \right) \quad (14)$$

where F is the satisfaction rate of the ecological suitable flow (%); $Q_{j,t,E}$ is the ecological suitable flow of the jth channel in the tth period (m^3/s), and $Q_{j,t,e}$ is the flow of the jth channel in the tth period (m^3/s).

3.1.2. Constraints

(1) Water balance constraint

The water balance constraint serves both as a model constraint and as the state variable transfer equation of the model

Please cite this article as: Li, J et al., Long-term impact of the Guxian Reservoir on water and sediment regulation in the middle reaches of the Yellow River Basin, International Journal of Sediment Research, https://doi.org/10.1016/j.ijsrc.2026.10.001

$$V_{i,t+1} = V_{i,t} + (Q_{in,i,t} - Q_{out,i,t})\Delta t \quad (15)$$

where $V_{i,t}$ and $V_{i,t+1}$ are the storage capacity of the ith reservoir at the initial and end of the t+1st period (m^3); $Q_{in,i,t}$ and $Q_{out,i,t}$ are average inflow and outflow of the ith reservoir in the tth period, respectively (m^3/s).

(2) Reservoir water level constraint

The upper limits of reservoir water levels include the flood control level during the flood season and the normal high water level during the non-flood season. The lower limits are dead water levels of reservoirs.

$$Z_{\min\ i,t} \le Z_{i,t} \le Z_{\max\ i,t} \quad (16)$$

where $Z_{i,t}$ is the water level of the ith reservoir at the end of the tth period (m); $Z_{\min\ i,t}$ and $Z_{\max\ i,t}$ are the lower and upper limit of the water level of the ith reservoir at the end of the tth period, respectively (m).

(3) Outflow constraint

The flow constraint includes the bankfull discharge limit and the basic ecological flow limit.

$$Q_{\min,i,t} \le Q_{out,i,t} \le Q_{\max,i,t} \quad (17)$$

where $Q_{\min,i,t}$ is the lower limit of the outflow of the ith reservoir in the tth period, which is ecological basic flow in the downstream of the ith reservoir (m^3/s); and $Q_{\max,i,t}$ is the upper limit of the outflow of the ith reservoir in the tth period, which is the bankfull discharge in the downstream of the ith reservoir (m^3/s).

(4) Power output constraint

$$N_{i,t} \le N_{install,i} \quad (18)$$

where $N_{i,t}$ is the power output of the ith reservoir in the tth period (kW); and $N_{install,i}$ is the installed capacity of the ith reservoir (kW).

(5) Storage constraints at the beginning and end of the period

Taking the actual water storage volumes of each reservoir at the beginning and end of each year under practical conditions as the constraints of the model.

$$\begin{cases} V_{i,0} = V_i^{begin} \\ V_{i,T} = V_i^{end} \end{cases} \quad (19)$$

where $V_{i,0}$ is the storage of the ith reservoir at the beginning of the period (m^3); V_i^{begin} is the practical storage of the ith reservoir at the beginning of the period (m^3); $V_{i,T}$ is the storage of the ith reservoir at the end of the period (m^3); and V_i^{end} is the practical storage of the ith reservoir at the end of the period (m^3). In this study, the water level of the GX Reservoir at the end of the flood season is generally maintained at or below the flood limit level, as specified in the *Feasibility Study Report of the Guxian Water Conservancy Project in the Yellow River Basin*.

3.1.3. Model assumptions and simplifications

The proposed LMSWSR model covers a 10-year operation period. Given the high-dimensional decision variables, constraints, and the significant computational complexity of the optimization algorithm, it is necessary to balance computational tractability with simulation accuracy. Since this study primarily focuses on the long-term balance of water and sediment in the middle reaches of the Yellow River, the following assumptions and simplifications were adopted.

(1) Simplification of sediment discharge rules of the SMX and XLD reservoirs

For the SMX and XLD reservoirs, the sediment discharge calculations during the flood season rely on empirical equations, rather than being derived from physically-based mechanistic models. In actual engineering practice, reservoir sediment flushing is carried out according to real-time water and sediment conditions, reservoir water levels, and other factors. However, to ensure computational feasibility and tractability for the long-series, multi-objective synergistic optimization framework developed in this study, the simplified empirical approach was adopted. While this simplification fulfills the requirements for long-term strategic evaluation, it inherently limits the model's capability to capture short-term sediment dynamic processes.

(2) Simplification of water-sediment routing and channel morphodynamics

As this study is positioned at the level of long-term water resources planning and regulation, integrating high-fidelity 2D or 3D hydrodynamic morphodynamic models into the optimization framework would exceed current computational limits. Therefore, the along-channel water and sediment evolution in the Xiaobei mainstream and the downstream channel were simplified. The empirical equations were used to calculate the total erosion and deposition volumes, and the inflow of downstream reservoirs was determined by direct runoff and sediment superposition, without simulating the refined spatial distribution characteristics of scouring and silting.

(3) Simplification of ecological objectives

The focus of the model is not on ecological scheduling. Consequently, the ecological objective was simplified to maximizing ecological suitable flow satisfaction rate, with the ecological basic flow strictly enforced as a boundary constraint. While some key ecological factors—such as aquatic organism habitats, sediment grain size dynamics, wetland ecology, and ecological pulse flows—are crucial for the Yellow River Basin, incorporating these factors requires specialized ecological data and a significantly finer temporal resolution. This would exponentially increase the computational difficulty. Thus, these detailed indicators were excluded from the current regulation model.

(4) Simplification of water supply objective for SMX and XLD reservoirs

Due to the difficulty in obtaining the actual dynamic water demand data for the specific intake areas of the SMX and XLD reservoirs, only their historical actual water supply data released by the Yellow River Conservancy Commission were accessible. Therefore, only the water supply satisfaction rate of the GX Reservoir was established as an optimization objective. For the SMX and XLD reservoirs, the model was simplified to ensure that their water supply volumes meet this actual water supply.

Please cite this article as: Li, J et al., Long-term impact of the Guxian Reservoir on water and sediment regulation in the middle reaches of the Yellow River Basin, International Journal of Sediment Research, https://doi.org/10.1016/j.ijsrc.2026.10.001

3.2. Model solving and scheme decision-making

3.2.1. Solving algorithm

Multi-objective optimization regulation of reservoirs typically involves multiple reservoirs, multiple objectives, and multiple constraints, which rendered the problem highly complex. The Non-dominated Sorting Genetic Algorithm II (NSGA-II) was proposed by Deb et al. (2002) and is currently recognized as one of the algorithms with the best effect in solving multi-objective problems. In this study, the NSGA-II was employed to solve the GX-SMX-XLD LMSWSR model.

3.2.2. Decision method

After a Pareto set was obtained by NSGA-II, the optimal solution still needed to be selected from the Pareto set to maximize the overall benefits. In this study, the comprehensive weights of indicators were determined using a combination of subjective and objective approaches. The subjective weights were derived using the Analytic Hierarchy Process (AHP) (Mandal et al., 2023), whereas the objective weights were calculated using the entropy weight method (Chen, 2021). The Technique for Order of Preference by Similarity to Ideal Solution (TOPSIS) was subsequently employed to evaluate the operation schemes of cascade reservoirs (Shih et al., 2007). In this manner, a comprehensive decision model for multi-objective regulation of cascade reservoirs was established.

3.3. Establishment of evaluation indicators

The construction of the GX Reservoir serves a threefold purpose: to strengthen the hydrodynamic force of water and sediment regulation of the XLD Reservoir, to scour and lower the elevation at the Tongguan (TG) station, and to improve the runoff utilization rate. A set of evaluation indicators was established to evaluate the comprehensive effects of the GX Reservoir. The regulation performance of the GX Reservoir was then analyzed by comparing model outputs under two scenarios: with and without the GX Reservoir.

(1) Assessment indicator for hydrodynamic force of water and sediment regulation of the XLD Reservoir

To prolong the service life of the sediment retention capacity of the XLD Reservoir, suitable water and sediment conditions are required to scour sediment deposits within the reservoir and reduce downstream channel sedimentation through water and sediment regulation. However, under current engineering conditions, the hydrodynamic force of water and sediment regulation of the XLD Reservoir remains insufficient. This deficiency is mainly reflected in two aspects: inadequate inflow flood dynamics and a limited number of days with high flow conducive to sediment discharge in the reservoir and sediment transport in the downstream reaches. One major function of the GX Reservoir is to counteract the severe attenuation of artificial flood peaks released from the Wanjiazhai (WJZ) Reservoir, thereby addressing the insufficient hydrodynamic force for water and sediment regulation of the XLD Reservoir. As the primary period for scouring in the XLD Reservoir is July, the reservoir inflow in July is employed as the key indicator for evaluating the hydrodynamic force of water and sediment regulation.

(2) Assessment indicator for the Tongguan Elevation

The Tongguan Elevation is influenced mainly by the water and sediment inflows of the Xiaobei mainstream and the Wei River, in addition to the operational mode of the SMX Reservoir. To examine the effects of the GX Reservoir operation on the Tongguan Elevation, these hydrological factors are considered. The calculation equation is expressed as follows (Hu et al., 2023):

$$Z_{TG} = -0.0061W_5 + 118.76\rho_5 + 327.74 \tag{20}$$

where Z_{TG} is the Tongguan Elevation at the end of the flood season in the middle reaches of the Yellow River (m); W_5 is the 5-year moving average of the total flow at LM, HX, ZT, and HJ stations (10^8 m^3); and ρ_5 is the 5-year moving average of the total sediment inflow coefficient at LM, HX, ZT, and HJ stations (kg·s/m^6).

(3) Assessment indicator for the runoff utilization rate

The Yellow River is characterized by water scarcity and high sediment load. A key challenge, therefore, is maximizing sediment transport while minimizing water use to improve runoff utilization efficiency. To quantify the role of the GX Reservoir in improving runoff utilization, the evaluation indicator is defined as the increase in sediment discharge into the sea relative to the total runoff during the regulation period, that is, the increment in sediment transport capacity.

$$\Delta\eta = \frac{1000*(S_2 - S_1)}{W_T} \tag{21}$$

where $\Delta\eta$ is the increment of sediment transport capacity, namely the increment of sediment flux per unit volume of water (kg/m^3); S_2 is the sediment discharge into the sea with the GX Reservoir during 10 years (10^8 t); S_1 is the sediment discharge into the sea without the GX Reservoir during 10 years (10^8 t); and W_T is the total runoff into the sea during 10 years (10^8 m^3).

4. Results and discussions

This section evaluates the performance of the proposed GX-SMX-XLD LMSWSR model and analyzes the comprehensive impacts of the GX Reservoir. First, to ensure the reliability and accuracy of the simulations, the established model is validated using historical observations. Following successful validation, the model is employed to conduct a 10-year series of optimal operations to assess the impacts of the GX Reservoir under various operational scenarios. Optimal operation is conducted on a monthly scale during the non-flood season and on a 10-day scale during the flood season. The population size of the optimization algorithm is set to 1000, and the number of iterations is 8000.

4.1. Model validation results

Empirical equations were applied to quantify the erosion and deposition volumes across the Xiaobei Mainstream, the SMX Reservoir, the XLD Reservoir, and the downstream channel, along with the Tongguan Elevation. The key parameters of these Equations were determined through a rigorous calibration process against historical data collected between 2001 and 2020. The detailed calibration outcomes are summarized below.

4.1.1. Verification of the empirical equations for the Xiaobei mainstream and downstream channel

The erosion and deposition volumes of the Xiaobei mainstream were calculated by Eqs. (5)–(7). The parameter calibration results of Eqs. (5) and (6) are shown in Fig. 4. As illustrated in the figures, the Nash-Sutcliffe Efficiency (NSE) for the flood season is 0.715, which falls between 0.5 and 0.75, indicating a good simulation

Please cite this article as: Li, J et al., Long-term impact of the Guxian Reservoir on water and sediment regulation in the middle reaches of the Yellow River Basin, International Journal of Sediment Research, https://doi.org/10.1016/j.ijsrc.2026.10.001

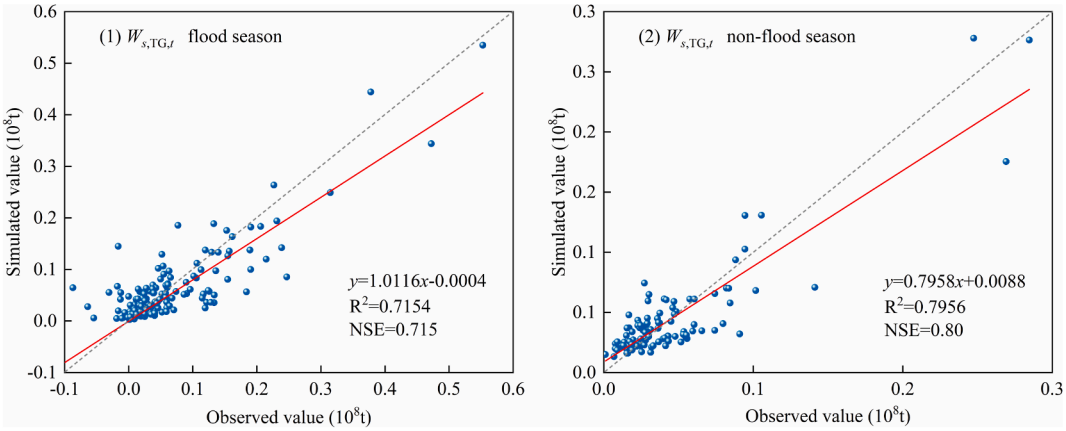


Fig. 4. Simulation results of sediment volume at TG Station during flood season and non-flood season (Subfigure (1) shows the simulation result of Eq. (5), and subfigure (2) shows the simulation result of Eq. (6). The gray dashed line represents the 1:1 reference line, while the red solid line denotes the linear regression line of the data).

performance; meanwhile, the NSE for the non-flood season is 0.80, exceeding 0.75, thereby demonstrating an excellent simulation performance. Overall, the simulation results of both equations exhibit high reliability, providing a solid scientific basis for the calculation of the sediment erosion and siltation amount in the Xiaobei mainstream using Eq. (7).

The erosion and deposition volumes of the downstream channel were calculated by Eqs. (8)–(11). The parameter calibration results of Eqs. (8)–(10) are shown in Fig. 5. As shown in the figure, during the flood season, the NSE is 0.677 when the XLD Reservoir releases sediment, indicating a good simulation performance; conversely, when there is no sediment release, the NSE reaches 0.769, demonstrating an excellent simulation performance. For the non-flood season, the NSE is 0.689, which also indicates a good simulation performance. Overall, the simulation results of the three equations exhibit high reliability, thereby providing a scientific basis for the calculation of the sediment erosion and siltation amount in the downstream channel using Eq. (11).

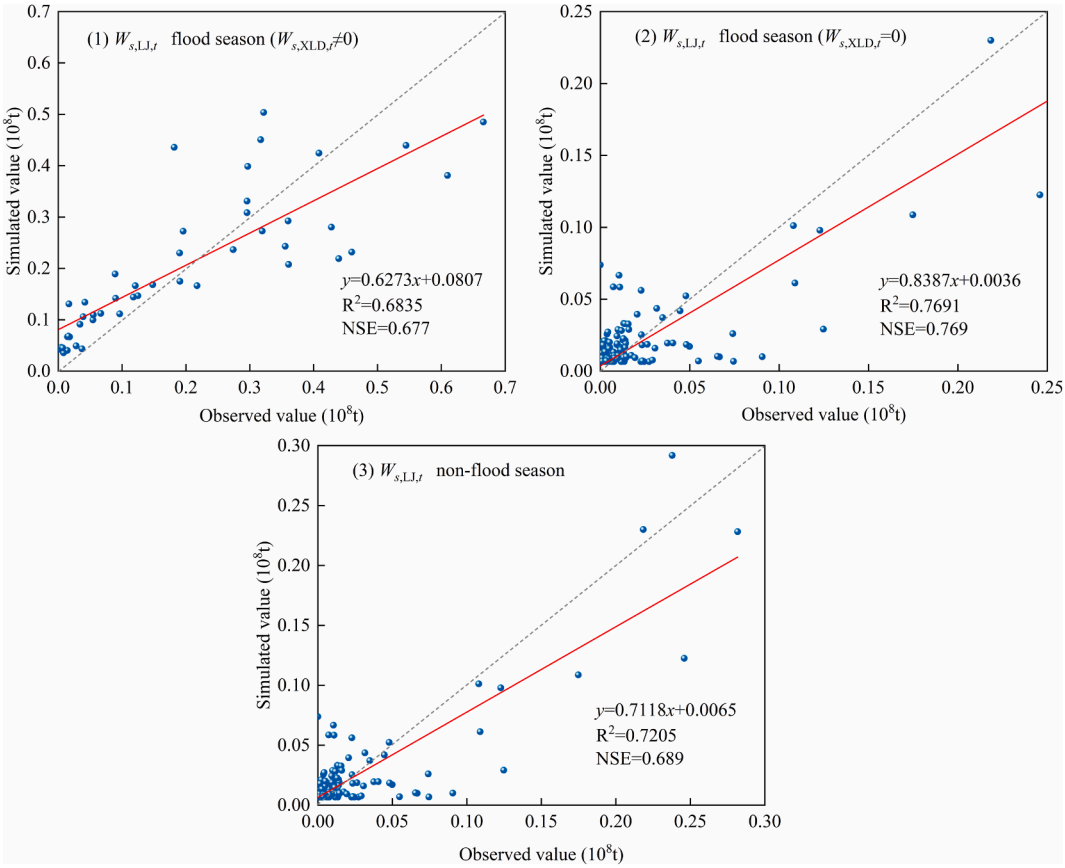


Fig. 5. Simulation results of sediment volume at LJ Station during flood season and non-flood season (Subfigures (1), (2), and (3) show the simulation results of Eqs. (8)–(10), respectively. The gray dashed line represents the 1:1 reference line, while the red solid line denotes the linear regression line of the data.).

Please cite this article as: Li, J et al., Long-term impact of the Guxian Reservoir on water and sediment regulation in the middle reaches of the Yellow River Basin, International Journal of Sediment Research, https://doi.org/10.1016/j.ijsrc.2026.10.001

4.1.2. Verification of the reliability of the optimal operation model

To verify the reliability of the optimization results from the GX-SMX-XLD LMSWSR model, a joint SMX-XLD optimization model was established. Adopting identical objectives, constraints, and parameter settings as the GX-SMX-XLD LMSWSR model, a continuous optimal operation was conducted over a three-year period comprising a wet year (2012), a normal year (2013), and a dry year (2014). The simulation results of the water levels and outflow sediment amount for the SMX and XLD reservoirs are presented in Fig. 6. As shown in the figure, the NSE values between the simulated and observed water levels and outflow sediment amounts for both the Sanmenxia and Xiaolangdi reservoirs all exceed 0.9, indicating an excellent goodness-of-fit. This demonstrates that the results of the joint SMX-XLD optimal operation model are highly reliable. Consequently, it further verifies the reliability of the GX-SMX-XLD LMSWSR model.

4.2. Impact of multi-objective synergistic water and sediment regulation

4.2.1. Reservoir and channel erosion

Fig. 7 presents the monthly erosion quantity of channels and reservoirs under different water and sediment series. As shown in the figure, the effect of the GX Reservoir on increasing the erosion quantity of the Xiaobei mainstream varies little among different water and sediment series. The increase in erosion quantity is similar in the high- and low-flow series, with increments of 2.51×10^8 t and 2.50×10^8 t, respectively, both slightly higher than the 2.41×10^8 t in the normal-flow series. The formation of this phenomenon can be explained as follows: under the low-flow series, the mean sediment runoff is large, the sediment retention effect of the GX Reservoir becomes the dominant factor for the increased erosion quantity of the Xiaobei mainstream; while under the high-flow series, the larger flow enhances the scouring effect during the flood season; under the normal-flow series, although the mean sediment runoff is similar to that of the high-flow series, the smaller flow results in a relatively weaker scouring enhancement effect.

The effects of the GX Reservoir on the erosion quantity of the SMX Reservoir followed the sequence "low-flow series > normal-flow series > high-flow series", with increases of 4.53×10^8 t, 1.72×10^8 t, and 0.03×10^8 t, respectively. The impact of the GX Reservoir on the erosion quantity of the SMX Reservoir during the flood season was limited. Its primary role was reflected in reducing the inbound sediment transport rate of the SMX Reservoir through sediment retention during the non-flood season. The mainstream sediment runoff itself exhibited the same pattern of low-flow series > normal-flow series > high-flow series, which reinforced this difference.

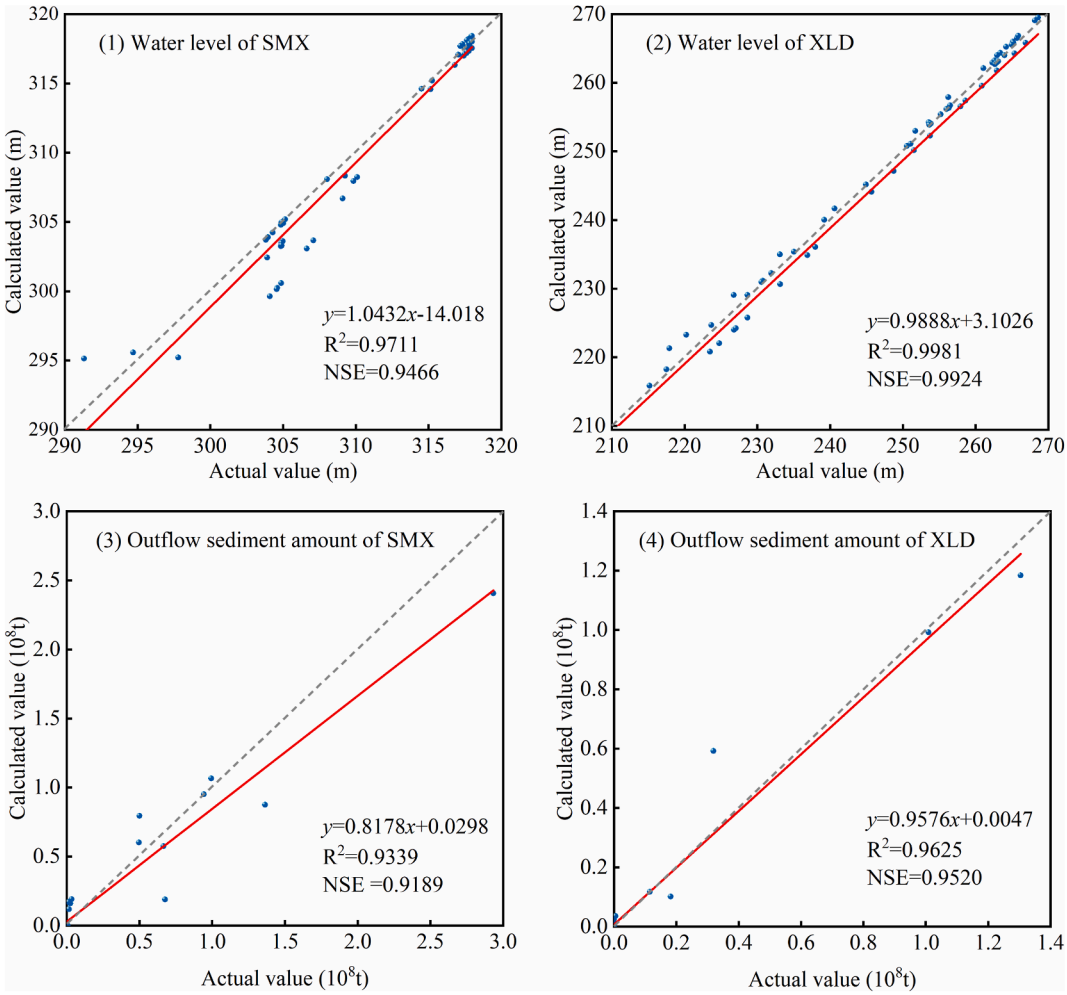


Fig. 6. Simulation results of the water levels and outflow sediment amount for the SMX and XLD reservoirs (The gray dashed line represents the 1:1 reference line, while the red solid line denotes the linear regression line of the data).

Please cite this article as: Li, J et al., Long-term impact of the Guxian Reservoir on water and sediment regulation in the middle reaches of the Yellow River Basin, International Journal of Sediment Research, https://doi.org/10.1016/j.ijsrc.2026.10.001

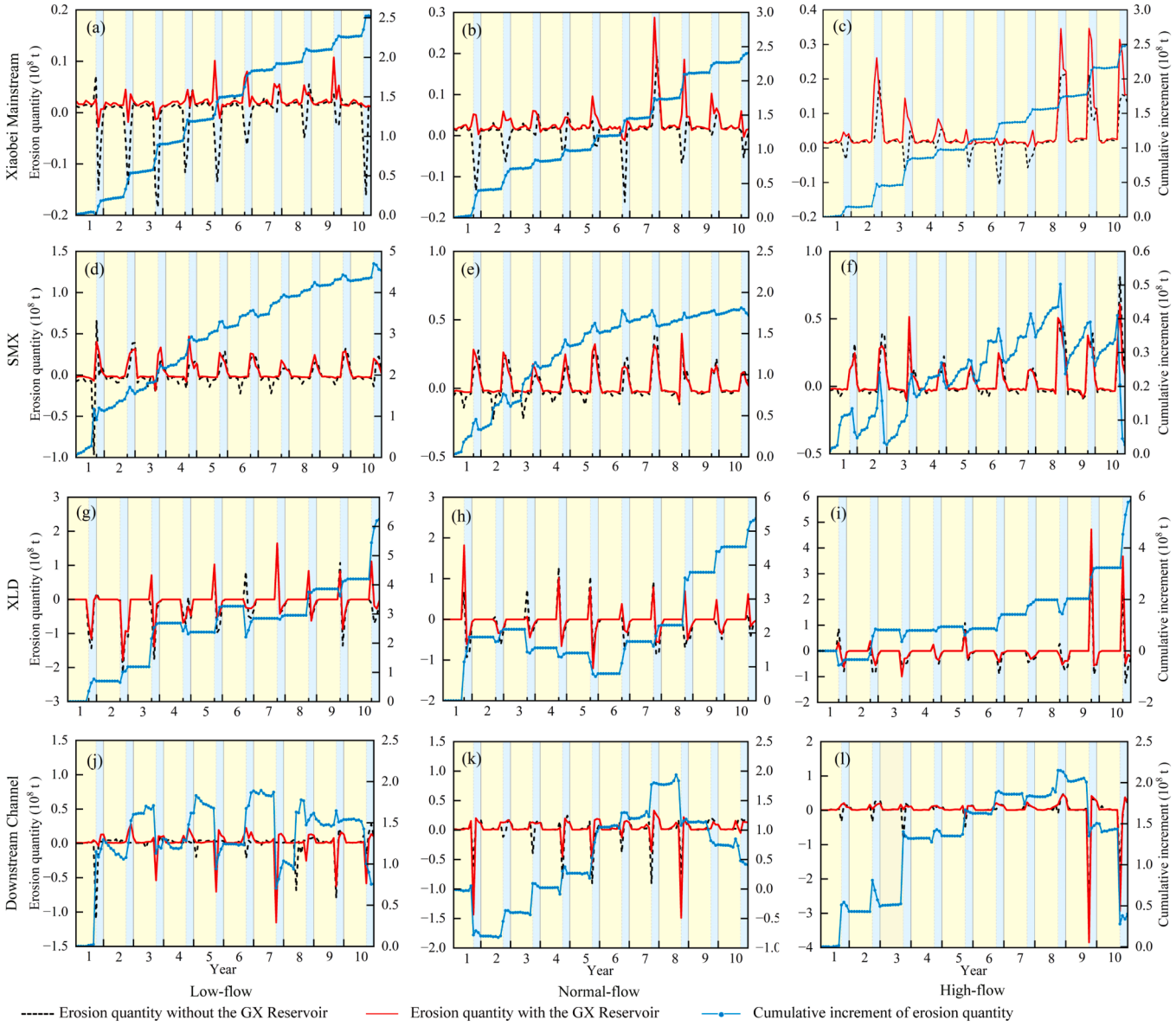


Fig. 7. Monthly erosion quantity of channels and reservoirs under different water and sediment series (A positive value "+" indicates erosion, and a negative value "-" indicates siltation. The yellow background indicates non-flood season, and the blue background indicates flood season).

The effects of the GX Reservoir on the erosion quantity of the XLD Reservoir followed the sequence "low-flow series > high-flow series > normal-flow series", with increases of 6.24 × 10^8 t, 5.83 × 10^8 t, and 5.38 × 10^8 t, respectively. In particular, the erosion quantity in the reservoir was enhanced during the early flood season, whereas sedimentation was reduced during the late flood season. The sediment reduction effect under the low-flow series was stronger than that under the other series. This is because the GX Reservoir significantly reduced the inbound sediment transport rate of the SMX Reservoir. This reduction led to a substantial decrease in the outbound sediment transport rate of the SMX Reservoir as well as the inbound sediment transport rate of the XLD Reservoir, which consequently reduced the sedimentation of the XLD Reservoir. The sediment reduction effect of the high-flow series is better than that of the normal-flow series. This is because the sediment release of the XLD Reservoir is mainly concentrated in the low-water-level sediment release stage in early July. The high-flow series endows the sediment release stage with stronger hydrodynamic force, resulting in a better sediment release effect and ultimately making the scouring effect of the XLD Reservoir more significant. Erosion was more effective in the high-flow series than in the normal-flow series. This result could be attributed to the fact that, although the outflow sediment load of the SMX Reservoir was larger during the high-flow series than in the normal-flow series, the sediment release of the XLD Reservoir was mainly concentrated in the water level drawdown and sediment discharge stages of the early flood season. The water and sediment series in the flood season provided stronger flow dynamics during this stage (lower water level and larger flow), thereby enhancing the sediment discharge efficiency and ultimately amplifying the scouring effect in the XLD Reservoir area.

As the scouring of the downstream channel contradicts the sediment release from the XLD Reservoir, the increase in the erosion quantity in the XLD Reservoir may result in greater sedimentation in downstream channels in certain years. Overall, the GX Reservoir exerted positive effects on downstream channel scouring,

Please cite this article as: Li, J et al., Long-term impact of the Guxian Reservoir on water and sediment regulation in the middle reaches of the Yellow River Basin, International Journal of Sediment Research, https://doi.org/10.1016/j.ijsrc.2026.10.001

following the sequence of "low-flow series > normal-flow series > high-flow series". The erosion quantity of the downstream channel increased by 0.75×10^8 t, 0.41×10^8 t, and 0.40×10^8 t in the low-, normal-, and high-flow series, respectively.

Fig. 8 presents the total erosion quantity of Xiaobei mainstream, SMX Reservoir, XLD Reservoir, and the downstream channel under three water and sediment series with and without the GX Reservoir. According to Fig. 8, in general, the erosion quantity of channels and reservoirs increased with runoff, whereas there are special phenomena in the SMX Reservoir and the downstream channel.

The erosion quantity of the SMX Reservoir was lower in the normal-flow series than in the low-flow series with the GX reservoir operation. This was primarily due to the stronger scouring effect of the Xiaobei mainstream in certain years in the normal-flow series with the GX reservoir operation. However, the total erosion quantity of the Xiaobei mainstream and the SMX Reservoir still exhibited a pattern of "normal-flow series > low-flow series".

The erosion quantity of the downstream channel was lower in the high-flow series than in the normal-flow series. This was attributed to a drastic reduction in the sedimentation quantity of the XLD Reservoir in the high-flow series, even though slight scouring occurred, which significantly increased the sediment release and inhibited the scouring of the downstream channel.

Regardless of the water and sediment series, the GX Reservoir had significant positive effects on the total erosion quantity of channels and reservoirs. Notably, the XLD Reservoir could cause scouring of downstream channels while ensuring slight scouring itself in the high-flow series.

4.2.2. Water supply satisfaction rate of the GX Reservoir

Fig. 9 presents the water supply satisfaction rate of the GX Reservoir under various water and sediment series. The water supply satisfaction rate of the GX Reservoir remained high after its completion, with satisfaction rates consistently above 85% across all water and sediment series. The water supply satisfaction rate increased with increasing runoff, reaching 87.47%, 91.38%, and 92.97% in the low-, normal-, and high-flow series, respectively, demonstrating its reliable performance in meeting water demand under varying runoff scenarios.

4.2.3. Power generation of the GX, SMX, and XLD reservoirs

Fig. 10 presents the power generation of reservoirs under various water and sediment series. It indicates that the power generation of all reservoirs increased with runoff, and the GX Reservoir further enhanced the power output of the SMX Reservoir and the XLD Reservoir through runoff regulation.

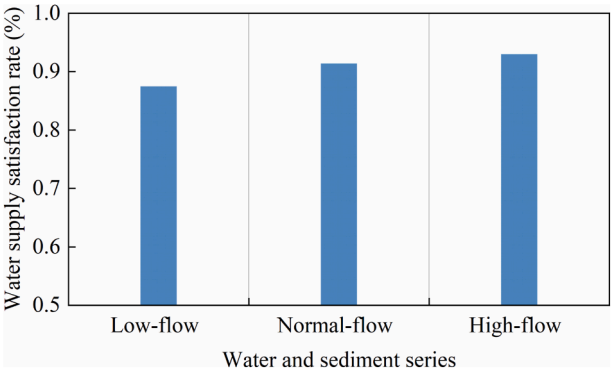


Fig. 9. Water supply satisfaction rate of GX Reservoir under different water and sediment series.

Under the joint operation of the GX, SMX, and XLD reservoirs, the power generation of the SMX Reservoir exhibited a relatively small increment. Specifically, its power generation reached 4.61×10^8 and 4.53×10^8 kW·h in the low- and normal-flow series, respectively. It increased slightly in the high-flow series, rising by 0.79×10^8 kW·h. The limited effect was mainly attributed to the relatively small installed capacity and restricted storage capacity of the SMX Reservoir, which constrained its regulation of runoff. Furthermore, in the absence of the GX Reservoir, the power generation flow of the SMX Reservoir during the flood season in the high-flow series was already close to its maximum power generation flow, thereby limiting the enhancement effect of the GX Reservoir.

Under the joint regulation of the GX, SMX, and XLD reservoirs, the power generation of the XLD Reservoir increased substantially with increasing runoff, with increments of 7.12×10^8, 22.87×10^8, and 30.33×10^8 kW·h in the low-, normal-, and high-flow series, respectively. The XLD Reservoir possessed a larger installed capacity and higher power generation flow, making it more adaptable to runoff variations. Its strong regulation capacity enabled the combined regulation with the GX Reservoir to fully exert its regulation effect, thereby further optimizing the power generation, especially in the high-flow series.

4.2.4. Ecological suitable flow satisfaction rate

Fig. 11 presents the ecological suitable flow satisfaction rate in the channels with and without the GX reservoir under various

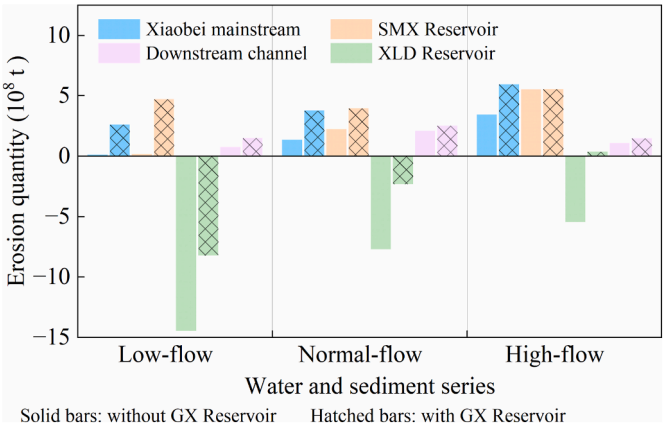


Fig. 8. Erosion quantity of channels and reservoirs under different water and sediment series.

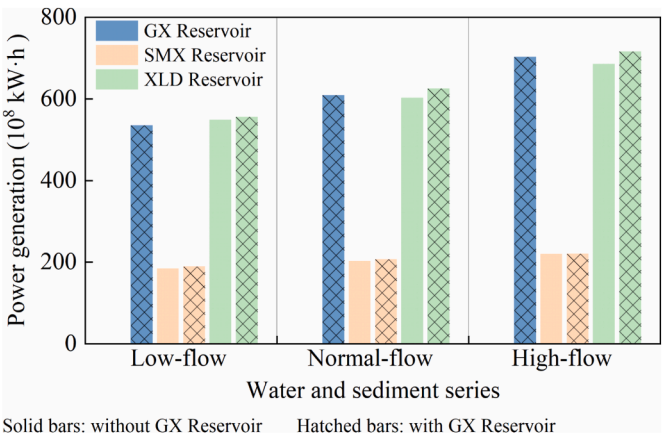


Fig. 10. Power generation of reservoirs under different water and sediment series.

Please cite this article as: Li, J et al., Long-term impact of the Guxian Reservoir on water and sediment regulation in the middle reaches of the Yellow River Basin, International Journal of Sediment Research, https://doi.org/10.1016/j.ijsrc.2026.10.001

water and sediment series. It illustrates that the ecological suitable flow satisfaction rate of the Xiaobei mainstream slightly increased with the increase of runoff with the GX Reservoir, reaching 98.42%, 98.69%, and 99% in low-, normal-, and high-flow series, respectively. In contrast, the ecological suitable flow satisfaction rate of the Xiaobei mainstream was slightly lower with the GX Reservoir than without it, with differences of 0.94%, 0.31%, and 0.63% in low-, normal-, and high-flow series, respectively. This outcome can be attributed to the influence of the GX Reservoir on water diversion patterns. In this study, the ecological suitable flow satisfaction rate of the Xiaobei mainstream was calculated as the ratio of the outflow at the Longmen hydrological station to the ecological suitable flow of the Xiaobei mainstream. Without the GX Reservoir, water was diverted directly from the mainstream and tributaries downstream of the Longmen station. After the GX Reservoir was put into operation, centralized water supply was adopted from the reservoir, indicating that water was diverted from upstream of the Longmen hydrological station. This led to a relative reduction in flow at Longmen station and, consequently, lowered the ecological suitable flow satisfaction rate.

With the GX Reservoir, the ecological suitable flow satisfaction rate of downstream channel increased with increasing of runoff, reaching 82.88%, 87.99%, and 89.59% in low-, normal-, and high-flow series, respectively. The increments in ecological suitable flow satisfaction rate of the downstream channel were 1.21%, 1.93%, and 1.02% in low-, normal-, and high-flow series, respectively.

4.3. Regulation performance

The operation of the GX Reservoir not only alters erosion and deposition patterns within the reservoirs and channels, water supply, power generation, and ecological conditions, but also impacts the hydrodynamic force of the XLD Reservoir, the Tongguan Elevation, and the runoff utilization rate. Consequently, sediment releasing ratios of 10%, 15%, and 20% were set for the GX Reservoir during the flood season to evaluate their effects on these indicators.

4.3.1. Effects on the hydrodynamic force of the XLD Reservoir

The inflows of the XLD Reservoir in July under different water and sediment series are shown in Fig. 12. As shown in the figure, under the high-, normal-, and low-flow series, the inflows for scenarios with the GX Reservoir (10%, 15%, and 20% sediment releasing ratios) are predominantly located above the curve of the without GX scenario. This indicates that the GX Reservoir can effectively increase the water volume entering the XLD Reservoir, thus providing more abundant hydrodynamic force for the water and sediment regulation. In certain years—such as the 8th year of the high- and normal-flow series and the 3rd year of the low-flow series—the overall inflow in July with the GX Reservoir was slightly lower than that without it. However, the GX Reservoir still increased the inflow of the XLD Reservoir in early- and mid-July, while also improving erosion quantity in the reservoir area. The specific impact characteristics under different inflow series are detailed as follows.

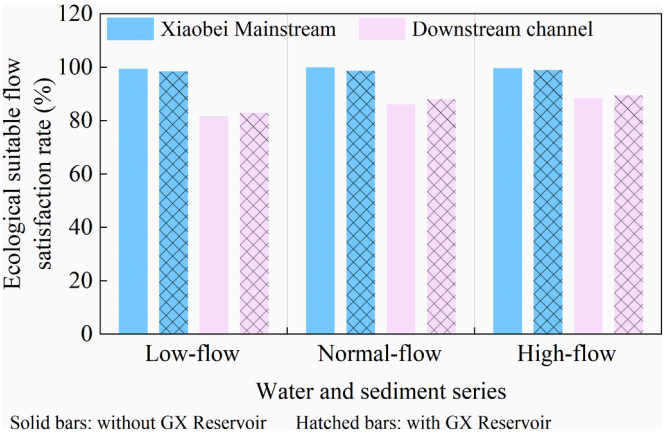


Fig. 11. Ecological suitable flow satisfaction rate of channels under different water and sediment series.

Under the high-flow series, the XLD Reservoir inflow in July fluctuated between 560 and 2760 m^3/s under the without GX scenario. The operation of the GX Reservoir further increases the July discharge. Particularly during periods with inherently large inflows, such as Years 2–3 and 9–10, the GX Reservoir can shape even larger flows. For instance, in the 9th year, the inflow under the without GX scenario was 2760 m^3/s, whereas the 10%, 15%, and 20% sediment releasing ratio schemes elevated the inflow to 3575 m^3/s, 3144 m^3/s, and 3820 m^3/s, respectively. Under this series, the inflow increment caused by the 20% sediment releasing ratio of the GX Reservoir ranged from 54 m^3/s to 1060 m^3/s. Overall, this represents a more substantial increase compared to the 10% and 15% sediment releasing ratios.

Under the normal-flow series, the XLD Reservoir inflow mostly ranged from 400 to 2210 m^3/s without the GX Reservoir. After the joint operation of GX-SMX-XLD reservoirs, the reshaping of the flow exhibited more severe fluctuations. When the sediment releasing ratio was set to 10%, the increment of the July inflow to the XLD Reservoir ranged from 176 m^3/s to 2206 m^3/s, showing the largest overall increase compared to the 15% and 20% ratios. However, the impacts of different sediment releasing ratios varied significantly interannually. For example, in the 3rd year, the 10% sediment releasing ratio scheme generated an exceptionally high flow peak (2661 m^3/s), which was much higher than those of the 15% (1083.5 m^3/s) and 20% (920.7 m^3/s) schemes in the same year; conversely, in the 7th year, the 20% sediment releasing ratio scheme released a larger discharge. This indicates that in normal-flow years, the setting of the sediment releasing ratio is more sensitive to its runoff and sediment processes, and the enhancing effect on hydrodynamic force presents stronger non-linear characteristics.

Under the low-flow series, the hydrodynamic force was deficient under the without GX scenario. The discharge dropped below 1000 m^3/s in most years, with a minimum of only 376 m^3/s, making it difficult to meet the basic requirements for initiating the water and sediment regulation. With the operation of the GX Reservoir, all three sediment releasing ratio schemes played a baseline-supporting role, stably maintaining the inflow discharge within the 1000–1500 m^3/s range for most years. Overall, the differences in the enhancing effects of varying sediment releasing ratios on the XLD hydrodynamic force were marginal during low-flow years. This is primarily because the available water resources are inherently limited under the low-flow series, requiring a trade-off between sediment flushing, water supply, and other operational objectives.

4.3.2. Effects on the Tongguan Elevation

Fig. 13 presents the calculation results of the Tongguan Elevation, with the initial value set at 326.75 m. As illustrated in the figure, under the high-, normal-, and low-flow series, the evolution curves of the Tongguan elevation for the scenarios with the GX Reservoir (10%, 15%, and 20% sediment releasing ratios) are consistently located below the without GX baseline curve. Regarding the internal differences among the various sediment

Please cite this article as: Li, J et al., Long-term impact of the Guxian Reservoir on water and sediment regulation in the middle reaches of the Yellow River Basin, International Journal of Sediment Research, https://doi.org/10.1016/j.ijsrc.2026.10.001

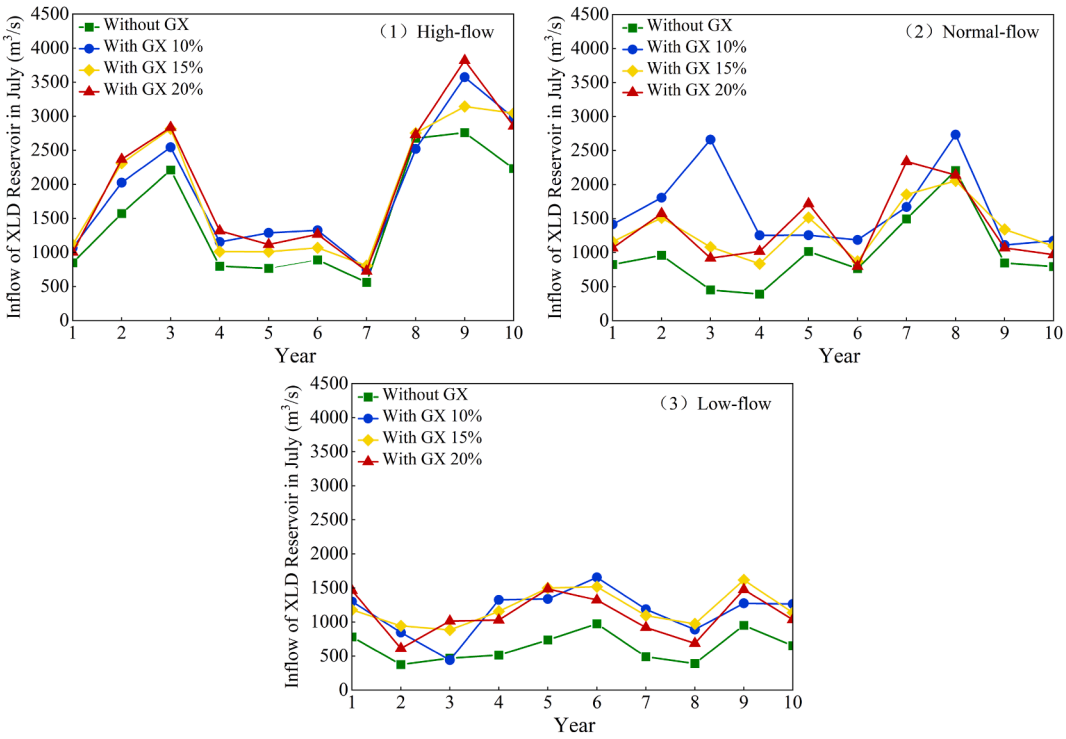


Fig. 12. Inflow of the XLD Reservoir in July under different water and sediment series.

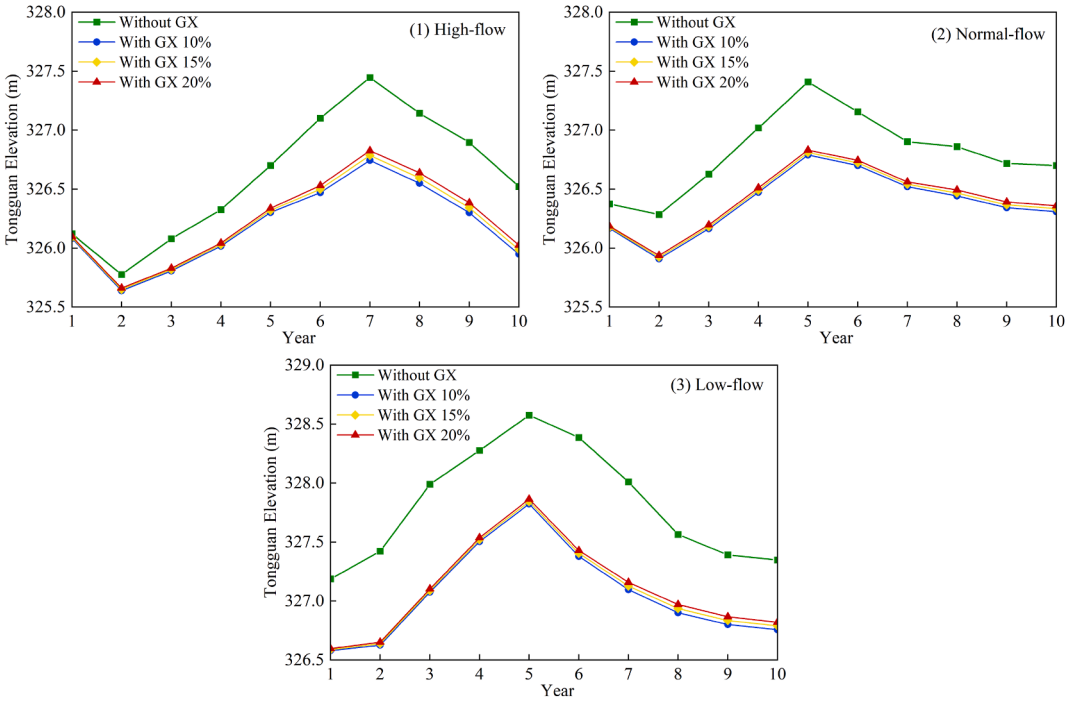


Fig. 13. The change of Tongguan Elevation under different water and sediment series.

releasing ratios of the GX Reservoir, their impacts on the Tongguan Elevation are generally marginal across the different water and sediment series. As the sediment releasing ratio decreases, the Tongguan elevation exhibits a slightly downward trend. This phenomenon indicates that a lower sediment releasing ratio at the GX Reservoir during the flood season is more conducive to reducing the Tongguan Elevation. However, because the Tongguan Elevation is comprehensively controlled by factors such as the operation mode of the SMX Reservoir and water and sediment conditions, relying solely on adjusting the sediment releasing ratio of the GX Reservoir offers limited potential for altering the Tongguan Elevation. Taking the 15% sediment releasing ratio as an

Please cite this article as: Li, J et al., Long-term impact of the Guxian Reservoir on water and sediment regulation in the middle reaches of the Yellow River Basin, International Journal of Sediment Research, https://doi.org/10.1016/j.ijsrc.2026.10.001

example, the variations in the Tongguan elevation under the high-, normal-, and low-flow series are as follows.

In the high-flow series, without the GX Reservoir, the Tongguan Elevation significantly decreased by 0.98 m in the first two years due to abundant runoff and low sediment runoff from tributaries. In the 3rd year, despite the high runoff at the four middle-reach stations, the sediment runoff at Huaxian Station caused the elevation rising to 326.33 m. In the 4th−7th years, with the reduced runoff, the elevation declined to 327.14 m. In the 8th−10th years, with rising runoff, the elevation decreased and stabilized at 326.52 m, with a 0.23 m cumulative reduction in ten years. After the operation of the GX Reservoir, the overall trend showed negligible changes, whereas the sediment retention effect reduced the magnitude of the variation. Specifically, the elevation decreased by 1.1 m in the first two years, rose to 326.03 m in the 3rd year, reached 326.79 m in the 4th−7th year, and stabilized at 325.99 m in the 8th−10th year, with a cumulative decrease of 0.76 m in ten years. Compared with the scenario without the GX Reservoir, the elevation was reduced by 0.53 m.

In the normal-flow series, without the GX Reservoir, the runoff at the four stations in the middle reaches was sufficient in the first two years, leading to a 0.47 m decrease in the Tongguan Elevation. In the 3rd−5th year, with reduced runoff, the elevation rose to 327.41 m. In the 6th−10th years, the elevation decreased and stabilized at 326.7 m, with a 0.05 m cumulative reduction in ten years. After the operation of the GX Reservoir, the overall trend remained consistent, whereas the sediment retention effect weakened the magnitude of both increases and decreases. Specifically, the elevation decreased by 0.83 m in the first two years, rose to 326.81 m in the 3rd−5th year, and finally stabilized at 326.33 m, with a cumulative reduction of 0.42 m in ten years. Compared with the scenario without the GX Reservoir, the elevation was reduced by 0.37 m.

In the low-flow series, without the GX Reservoir, the Tongguan Elevation continuously increased to 328.58 m in the first five years of the regulation period, primarily due to low runoff and high sediment runoff at the four stations in the middle reaches. In the 6th−10th years, due to higher runoff and reduced sediment runoff, the Tongguan Elevation decreased and stabilized at 327.35 m, with a 0.6 m cumulative rise in ten years. After the operation of the GX Reservoir, the Tongguan Elevation decreased by 0.16 m in the first year, as relatively sufficient runoff and sediment runoff interception by the reservoir reduced most of the sediment input. In the 2nd−5th year, the elevation showed an upward trend, but the magnitude was smaller than that without the GX Reservoir. In the 6th−10th year, under the improved runoff in the middle reaches, the elevation stabilized at 326.79 m, with a 0.04 m cumulative increase in ten years. Overall, the GX Reservoir reduced the Tongguan Elevation by 0.56 m in the low-flow series.

Overall, with or without the GX Reservoir, the Tongguan Elevation decreases significantly in response to an increase in water or a decrease in sediment. However, when the flood-season sediment releasing ratio of the GX Reservoir is set at 15%, compared to the without GX scenario, it effectively suppresses the rise of the Tongguan Elevation under the low-flow series, achieving the most significant reduction of 0.56 m. This is primarily because the sediment from the mainstream is substantial during the low-flow series, and the GX Reservoir exerts a remarkable interception effect on this massive sediment load. Furthermore, the Tongguan Elevation is further lowered under the normal- and high-flow series. Although the sediment in these two series is comparable, the high-flow series yields a more pronounced reduction (0.53 m) compared to the normal-flow series (0.37 m) due to its more abundant water supply.

4.3.3. Effects on runoff utilization rate

Fig. 14 presents the sediment discharge into the sea and sediment transport capacity in different water and sediment series. Overall, under the high-, normal-, and low-flow series, the GX Reservoir can significantly enhance the sediment transport capacity in the lower reaches of the Yellow River and substantially increase the sediment discharge into the sea.

Under the high-flow series, the sediment discharges into the sea for the scenarios with the GX Reservoir all exceed 21.0×10^8 t, achieving a substantial increase compared to the without GX scenario (14.19×10^8 t). In this series, the 15% sediment releasing ratio scenario exhibits the optimal sediment transport efficiency, with the increment of sediment transport capacity reaching the maximum value for this series (3.72 kg/m^3). This indicates that during high-flow periods, a 15% flood-season sediment releasing ratio can optimally couple with the abundant hydrodynamic forces, thereby achieving efficient sediment flushing and channel scouring in the downstream reaches.

Under the normal-flow series, the sediment discharge into the sea for the without GX scenario is only 11.29×10^8 t. After implementing operations with 10%, 15%, and 20% sediment releasing ratios, the sediment discharge into the sea increases to 18.43×10^8t, 18.05×10^8 t, and 16.93×10^8 t, respectively. Correspondingly, the increment of sediment transport capacity presents an increasing trend as the sediment releasing ratio decreases, with the 10% sediment releasing ratio scenario achieving the maximum increment (3.93 kg/m^3). This demonstrates that under normal-flow conditions, a lower sediment releasing ratio can better coordinate the flow-sediment relationship, shaping a more favorable hydrodynamic process for sediment transport in the downstream channel.

Under the low-flow series, due to deficient overall hydrodynamics, the overall improvement in sediment transport capacity is significantly weaker than that in the high- and normal-flow series. Even so, the regulation of the GX Reservoir still plays a crucial supportive role, elevating the sediment discharge into the sea from 14.88×10^8 t without GX to a range of 16.00×10^8 t to 18.21×10^8 t. In this series, the 10% sediment releasing ratio scenario corresponds to the maximum sediment discharge (18.21×10^8 t) and transport increment (2.05 kg/m^3). This reflects that under water scarcity conditions where available water resources are limited, a lower sediment releasing ratio is more advantageous for concentrating the limited water volume to transport sediment into the sea.

4.4. Limitations of this study

While the proposed LMSWSR model provides valuable insights into the joint operation of cascade reservoirs, several simplifications were made in the model construction. First, the sediment discharge rules were simplified. The GX Reservoir was set to completely intercept sediment during the non-flood season and discharge at fixed ratios (e.g., 15%) during the flood season. The sediment discharge of the SMX and XLD Reservoirs during the flood season was calculated using empirical equations. This fixed-rule assumption differs from the dynamic operation logic of actual engineering, which is adjusted in real-time based on water-sediment conditions. Second, the spatial sedimentation and erosion simulation in reservoirs and channels were simplified, a lumped calculation method was adopted for reservoir and channel scouring and silting. This method only accounts for the total volume variations and fails to capture the spatial distribution characteristics of sedimentation and erosion across different river cross-sections. Finally, the ecological objective was evaluated solely using the ecological suitable flow satisfaction rate. Key

Please cite this article as: Li, J et al., Long-term impact of the Guxian Reservoir on water and sediment regulation in the middle reaches of the Yellow River Basin, International Journal of Sediment Research, https://doi.org/10.1016/j.ijsrc.2026.10.001

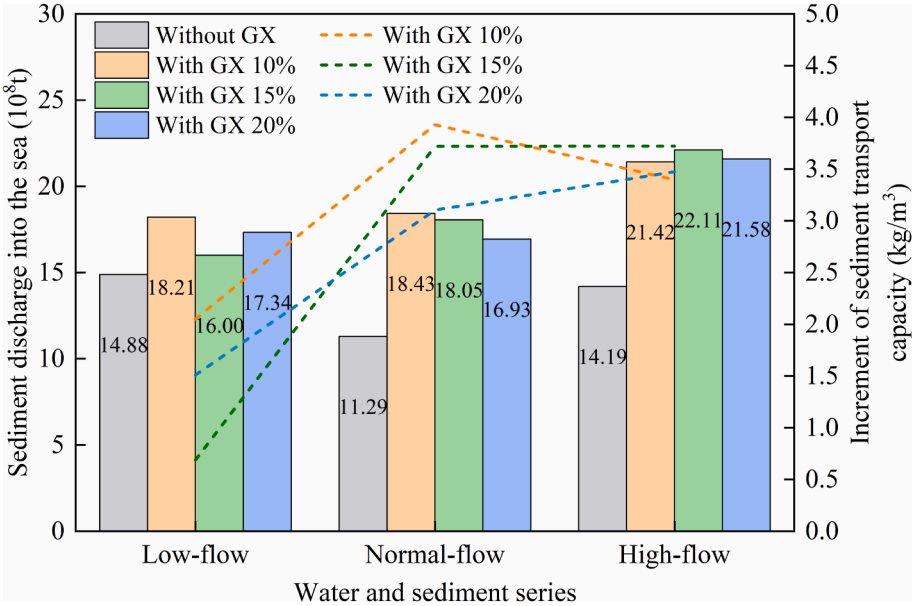


Fig. 14. Sediment discharge into the sea and the sediment transport capacity under different water and sediment series (The bar charts represent the sediment discharge into the sea, and the line graphs represent the increment of sediment transport capacity.).

ecological dimensions of the Yellow River Basin, such as aquatic organism habitats, wetland ecosystems, and specific sediment grain size requirements, were not considered. These limitations highlight the need for more complex, physically based dynamic models and comprehensive multi-dimensional ecological indexes in future studies.

5. Conclusions

To address water and sediment issues in the middle and lower reaches of the Yellow River, the LMSWSR model for GX, SMX, and XLD reservoirs was established, incorporating the objectives of reservoir and channel desilting, water supply, power generation, and ecological protection. The comprehensive regulatory role of the GX Reservoir was analyzed under high-, normal-, and low-flow series. The main conclusions are as follows.

(1) Across all water and sediment series, the GX Reservoir had positive effects on reservoir and channel scouring, power generation, and the ecology of downstream channels in the early sediment detention stage, while the ecological suitable flow satisfaction rate of the Xiaobei mainstream decreased slightly with the GX Reservoir due to altered water supply conditions.
(2) The implementation of different sediment releasing ratio schemes at the GX Reservoir has significantly enhanced the inflow to the XLD Reservoir in July across all water and sediment series, thereby boosting the hydrodynamic force for the water and sediment regulation.
(3) The joint operation with the GX Reservoir effectively reduces the Tongguan Elevation across all water and sediment series. However, adjusting the flood-season sediment releasing ratio (10%–20%) exerts only a marginal influence.
(4) The GX Reservoir can significantly enhance the sediment transport capacity of the downstream channel under different sediment releasing ratios. A 15% ratio maximizes sediment transport efficiency in high-flow years, while a lower ratio of 10% proves most advantageous in normal- and low-flow years.

This study established the high-, normal-, and low-flow series based on historical data and assessed the regulation performance of the GX Reservoir during its initial operational decade. Nevertheless, the present study is subject to several methodological limitations, including the reliance on fixed rather than dynamic sediment discharge rules, the lumped calculation of channel scouring that omits spatial physical routing, and the use of a single ecological index. In the context of global climate change, future research should integrate fully dynamic physical models and evaluate the cumulative effects of cascade water and sediment regulation over the entire lifecycle of the GX Reservoir.

CRediT authorship contribution statement

Jieyu Li: Writing – original draft, Methodology, Funding acquisition, Formal analysis. **Lei Li:** Visualization, Software, Formal analysis. **Donglin Li:** Writing – original draft, Funding acquisition. **Gang Liu:** Validation, Software. **Guoqing Sang:** Writing – review & editing, Funding acquisition. **Yuanjian Wang:** Writing – review & editing, Supervision, Methodology, Funding acquisition.

Data availability

Data will be made available on request.
Yellow River Conservancy Commission, 2001–2020.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This research was funded by the National Natural Science Foundation of China (Grant No. U2243601, 52409090), Taishan Scholars Young Experts Program (Grant No. tsqnz20250744), Key Research & Development Program of Shandong Province, China (Grant No. 2024TSGC0154), Science and Technology Program of

University of Jinan (Grant No. XRC2560), Zhongyuan Science and Technology Innovation Leading Talents Program (Grant No. 254200510046); Special Project for Basic Research Business Fees of the Yellow River Institute of Hydraulic Research (HKY-JBYW-2024-01).

References

Belletti, B., de Leaniz, C. G., Jones, J., Bizzi, S., Börger, L., Segura, G., Castelletti, A., van de Bund, W., Aarestrup, K., van de Bund, W., Barry, J., Belka, K., Berkhuysen, A., Birnie-Gauvin, K., Birnie-Bussettini, M., Birnie-Gauvin, K., Birnie-Carolli, M., Birnie-Consuegra, S., Dopico, E., Feierfeil, T.Fernández, S., … (2020). More than one million barriers fragment Europe's rivers. *Nature, 588*(7838), 436–441, https://doi.org/10.1038/s41586-020-3005-2

Canter, L. W., & Kamath, J. (1995). Questionnaire checklist for cumulative impacts. *Environmental Impact Assessment Review, 15*(4), 311–339, https://doi.org/10.1016/0195-9255(95)00010-C

Chen, J., Zhou, W., Han, S., & Sun, G. (2017). Influences of retrogressive erosion of reservoir on sedimentation of its downstream river channel—A case study on Sanmenxia Reservoir and the Lower Yellow River. *International Journal of Sediment Research, 32*(3), 373–383, https://doi.org/10.1016/j.ijsrc.2017.02.007

Chen, P. (2021). Effects of the entropy weight on TOPSIS. *Expert Systems with Applications, 168*, 114186, https://doi.org/10.1016/j.eswa.2020.114186

Deb, K., Pratap, A., Agarwal, S., & Meyarivan, T. A. M. T. (2002). A fast and elitist multiobjective genetic algorithm: NSGA-II. *IEEE Transactions on Evolutionary Computation, 6*(2), 182–197, https://doi.org/10.1109/4235.996017

Dos Santos, N. C. L., Garcia-Berthou, E., Dias, J. D., Lopes, T. M., Affonso, I. D. P., Severi, W., & Agostinho, A. A. (2018). Cumulative ecological effects of a Neotropical reservoir cascade across multiple assemblages. *Hydrobiologia, 819*(1), 77–91, https://doi.org/10.1007/s10750-018-3630-z

Guo, C., Jin, Z., Guo, L., Lu, J., Ren, S., & Zhou, Y. (2020). On the cumulative dam impact in the upper Changjiang River: Streamflow and sediment load changes. *Catena, 184*, 104250, https://doi.org/10.1016/j.catena.2019.104250

Guo, C., Jin, Z., Liang, D., Nones, M., & Zhou, Y. (2024). Spatiotemporal evolution and influencing factors of flood control storage sedimentation in the Three Gorges Reservoir, Changjiang River, China. *Catena, 243*, 108214, https://doi.org/10.1016/j.catena.2024.108214

Habets, F., Molénat, J., Carluer, N., Douez, O., & Leenhardt, D. (2018). The cumulative impacts of small reservoirs on hydrology: A review. *Science of the Total Environment, 643*, 850–867, https://doi.org/10.1016/j.scitotenv.2018.06.188

Hou, C., Yi, Y., Song, J., & Zhou, Y. (2021). Effect of water-sediment regulation operation on sediment grain size and nutrient content in the lower Yellow River. *Journal of Cleaner Production, 279*, 123533, https://doi.org/10.1016/j.jclepro.2020.123533

Hu, C., Zhang, Z., & Zhang, X. (2023). Threshold system of regulation indicators for maintaining the runoff and sediment balance of the Yellow River basin. *Advances in Water Science, 34*(5), 647–659. *(in Chinese)*

Jiang, E., Qu, B., Jia, J., Liu, C., Wang, Y., Li, D., & Li, J. (2023). Study on the coordination of flow and sediment transport, ecological environment, and crop production in the Lower Yellow River since 2000. *River, 2*(3), 326–335, https://doi.org/10.1002/rvr2.61

Jin, W., Wang, Y., Chang, J., & Wang, X. (2024). Response of water resources multi-objective interests in middle-lower Yellow River to Guxian Reservoir operation. *Advances in Science and Technology of Water Resources, 44*(1), 52–59. *(in Chinese)*

Li, D., Chen, Y., Lyu, L., & Cai, X. (2024a). Uncovering historical reservoir operation rules and patterns: Insights from 452 large reservoirs in the contiguous United States. *Water Resources Research, 60*(8), e2023WR036686, https://doi.org/10.1029/2023WR036686

Li, F., Wan, Z., & Guo, B. (2021). *Feasibility Study report of the Guxian water conservancy project in the yellow River Basin*. Zhengzhou: Yellow River Engineering Consulting Co., Ltd.

Li, J., Li, H., Wang, Y., Liu, G., Yang, F., Wang, Q., & Li, D. (2024b). Multi-objective synergy and competition relationship of water and sediment regulation of the Heishanxia reservoir in upstream of the Yellow River Basin. *Journal of Hydrology: Regional Studies, 56*, 102045, https://doi.org/10.1016/j.ejrh.2024.102045

Li, L., Chen, L., Chen, S., Zhang, Y., Xu, Y., Zhi, X., & Tang, L. (2024c). The cumulative effects of cascade reservoirs control nitrogen and phosphorus flux: Base on biogeochemical processes. *Water Research, 252*, 121177, https://doi.org/10.1016/j.watres.2024.121177

Liu, G., Jiang, E., Li, D., Li, J., Wang, Y., Zhao, W., & Yang, Z. (2025). Annual multi-objective optimization model and strategy for scheduling cascade reservoirs on the Yellow River mainstream. *Journal of Hydrology, 659*, 133306, https://doi.org/10.1016/j.jhydrol.2025.133306

Liu, W., Wang, S., Sang, Y. F., Ran, L., & Ma, Y. (2021). Effects of large upstream reservoir operations on cross-sectional changes in the channel of the lower Yellow River reach. *Geomorphology, 387*, 107768, https://doi.org/10.1016/j.geomorph.2021.107768

Mandal, P., Mukhopadhyay, A., Mukherjee, S., Mukhopadhyay, U., Bhattacharya, S., & Paul, S. (2023). Assessing the suitability of a reservoir site in a water-stressed area in Purulia, India, using AHP-TOPSIS. *Sustainable Water Resources Management, 9*(3), 74, https://doi.org/10.1007/s40899-023-00852-z

Ministry of Water Resources of China. (2001–2020). *Hydrological yearbook of China: Yellow River Basin*. Beijing: Hydrology Bureau, MWR.

Mu, J., Ding, S., Liu, S. M., Song, G., Ning, X., Zhang, X., & Zhang, H. (2024). Multiple isotopes decipher the nitrogen cycle in the cascade reservoirs and downstream in the middle and lower Yellow River: Insight for reservoir drainage period. *Science of the Total Environment, 918*, 170625, https://doi.org/10.1016/j.scitotenv.2024.170625

Shih, H. S., Shyur, H. J., & Lee, E. S. (2007). An extension of TOPSIS for group decision making. *Mathematical and Computer Modelling, 45*(7–8), 801–813, https://doi.org/10.1016/j.mcm.2006.03.023

Sun, Y., Li, J., Cao, Z., & Borthwick, A. G. (2022). Vertically layered flow structure at confluence of a reservoir and tributary carrying high sediment loads. *Frontiers in Earth Science, 10*, 924005, https://doi.org/10.3389/feart.2022.924005

Timpe, K., & Kaplan, D. (2017). The changing hydrology of a dammed Amazon. *Science Advances, 3*(11), e1700611, https://doi.org/10.1126/sciadv.1700611

Wan, Z., Liu, J., & Li, F. (2013). Research of joint operation method of Guxian and Xiaolangdi Reservoirs. *Yellow River, 35*(10), 36–39. *(in Chinese)*

Wang, K., Huang, L., He, G., Fang, H., Chen, M., Wang, D., & Wu, X. (2023). Spatial-temporal evolution of sediment transport in the upper Yangtze River Basin considering the cumulative impacts of mega reservoirs. *Catena, 232*, 107370, https://doi.org/10.1016/j.catena.2023.107370

Wang, J., Walter, B. A., Yao, F., Song, C., Ding, M., Maroof, A. S., … Wada, Y. (2021). GeoDAR: Georeferenced global dam and reservoir dataset for bridging attributes and geolocations. *Earth System Science Data Discussions*, 1–52, https://doi.org/10.5194/essd-14-1869-2022, 2021.

Wang, G., Wu, B., & Wang, Z. Y. (2005). Sedimentation problems and management strategies of Sanmenxia Reservoir, Yellow River, China. *Water Resources Research, 41*(9), W09417, https://doi.org/10.1029/2004WR003919

Wang, Y., Zhang, N., Wang, D., Wu, J., & Zhang, X. (2018). Investigating the impacts of cascade hydropower development on the natural flow regime in the Yangtze River, China. *Science of the Total Environment, 624*, 1187–1194, https://doi.org/10.1016/j.scitotenv.2017.12.212

Wu, B., Wang, G., Wang, Z., & Xia, J. (2004). Effect of changes in flow runoff on the elevation of Tongguan in Sanmenxia Reservoir. *Chinese Science Bulletin, 49*(15), 1658–1664, https://doi.org/10.1007/BF03184139

Wu, X., Yue, Y., Borthwick, A. G., Slater, L. J., Syvitski, J., Bi, N., … Wang, H. (2023). Mega-reservoir regulation: A comparative study on downstream responses of the Yangtze and Yellow rivers. *Earth-Science Reviews, 245*, 104567, https://doi.org/10.1016/j.earscirev.2023.104567

Xia, J., Cheng, Y., Zhou, M., Yu, X., Xu, X., Blanckaert, K., & Wang, Z. (2025). Relation between bank erosion and bed incision in the braided reach of the Lower Yellow River undergoing channel degradation. *International Journal of Sediment Research, 40*(2), 241–252, https://doi.org/10.1016/j.ijsrc.2024.12.002

Yellow River Conservancy Commission. (2001–2020) [dataset]. *Yellow River water resources bulletin*. Retrieved from. http://yrcc.gov.cn/gzfw/szygb/index.html.

Yellow River Conservancy Commission. (2013). *Comprehensive Planning for the Yellow River Basin (2012–2030)*. Zhengzhou: The Yellow River Water Conservancy PR.

Zhang, J., Li, F., Lv, Q., Wang, Y., Yu, J., Gao, Y., & Lv, Z. (2021a). Impact of the water-sediment regulation scheme on the phytoplankton community in the Yellow River estuary. *Journal of Cleaner Production, 294*, 126291, https://doi.org/10.1016/j.jclepro.2021.126291

Zhang, J., Shang, Y., Liu, J., Lu, J., Wei, S., Wan, Z., & Fu, J. (2021b). Optimisation of reservoir operation mode to improve sediment transport capacity of silt-laden rivers. *Journal of Hydrology, 594*, 125951, https://doi.org/10.1016/j.jhydrol.2020.125951

Zhou, X., Huang, X., Zhao, H., & Ma, K. (2020). Development of a revised method for indicators of hydrologic alteration for analyzing the cumulative impacts of cascading reservoirs on flow regime. *Hydrology and Earth System Sciences, 24*(8), 4091–4107, https://doi.org/10.5194/hess-24-4091-2020

Please cite this article as: Li, J et al., Long-term impact of the Guxian Reservoir on water and sediment regulation in the middle reaches of the Yellow River Basin, International Journal of Sediment Research, https://doi.org/10.1016/j.ijsrc.2026.10.001